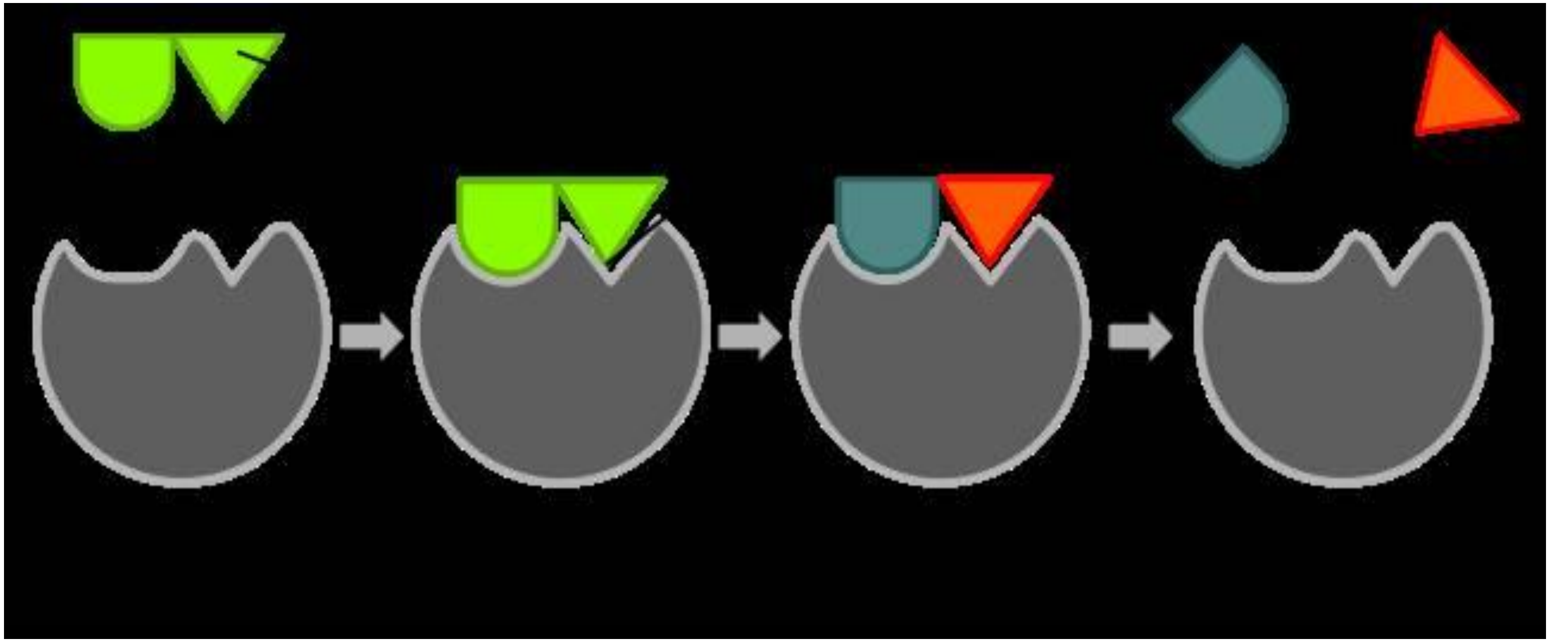
ENZYMES & METABOLISM

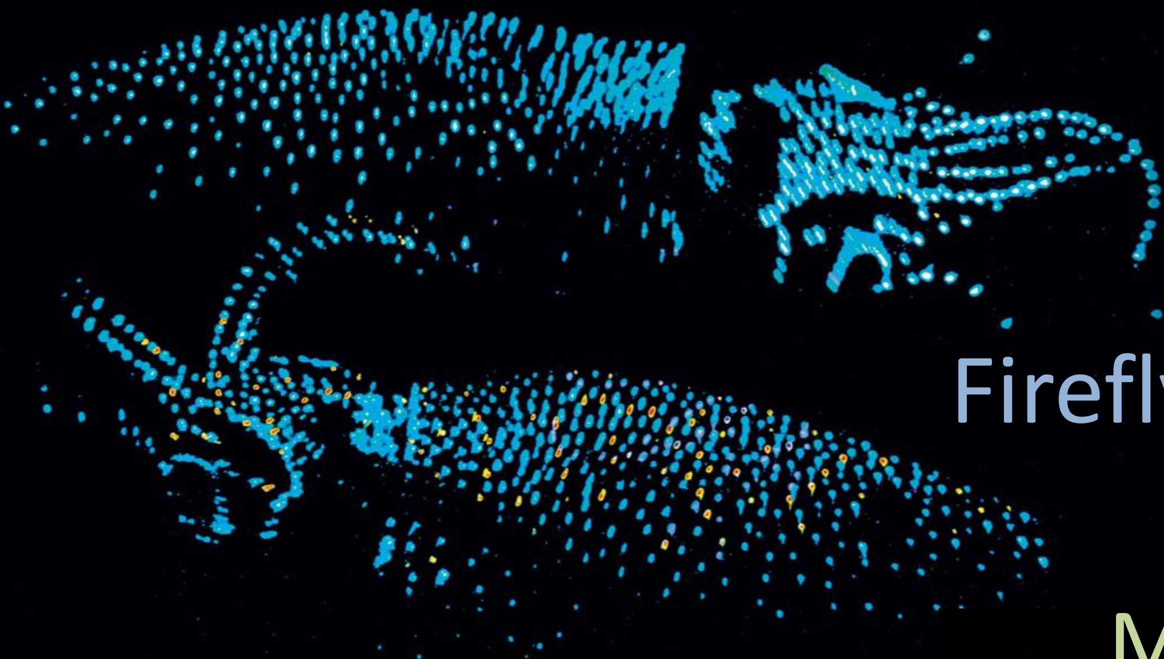


WEEK 2

Overview: The Energy of Life

- The living cell is a miniature chemical factory where thousands of reactions occur
- The cell extracts energy and applies energy to perform work
- Some organisms even convert energy to light, as in bioluminescence

Bioluminescence



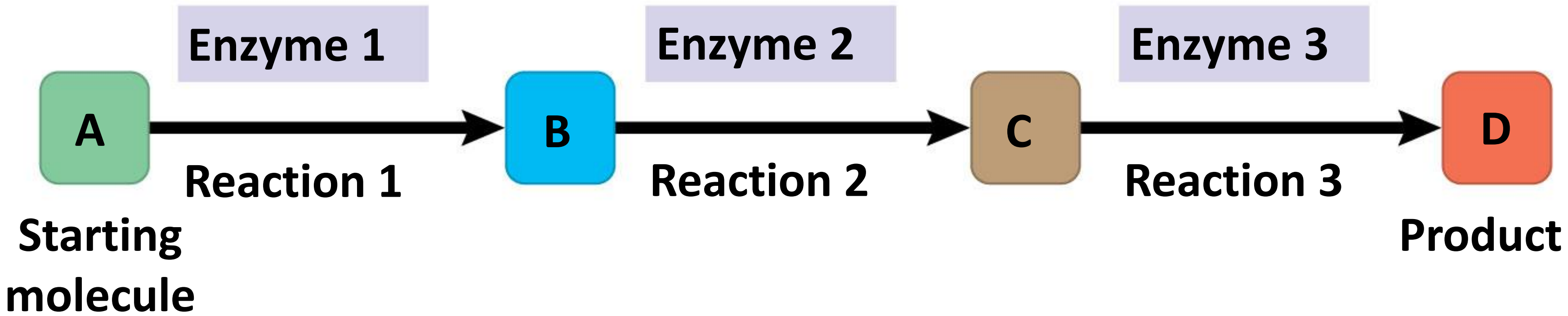
Firefly squid

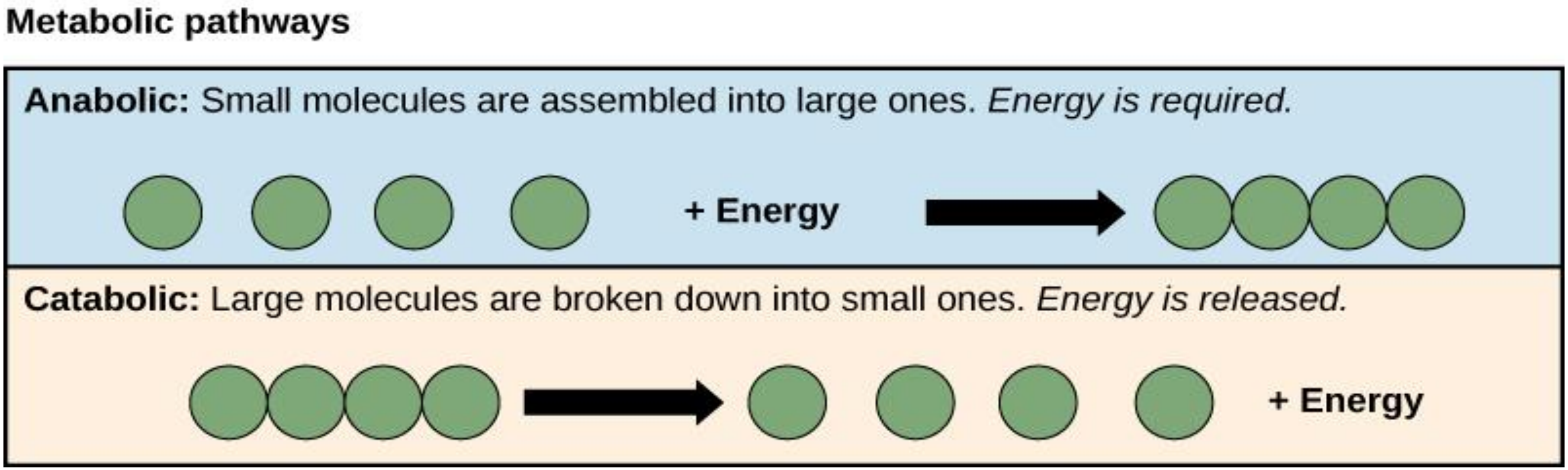
Mycena mushroom

Metabolism and Thermodynamics

- **Metabolism:** totality of an organism's chemical reactions
- An organism's metabolism transforms matter and energy, **subject to the <u>laws of thermodynamics</u>**
- Emergent property of life that arises from interactions between molecules within the cell

Metabolic pathway





- **Catabolic pathways** release energy by breaking down complex molecules into simpler compounds
- E.g. Cellular respiration, the breakdown of glucose in the presence of oxygen

- **Anabolic pathways** consume energy to build complex molecules from simpler ones
- E.g. The synthesis of protein from amino acids

Energy

- **Energy** is the capacity to cause change
- Energy exists in various forms, some of which can perform work
- Energy can be converted from one form to another

Forms of energy

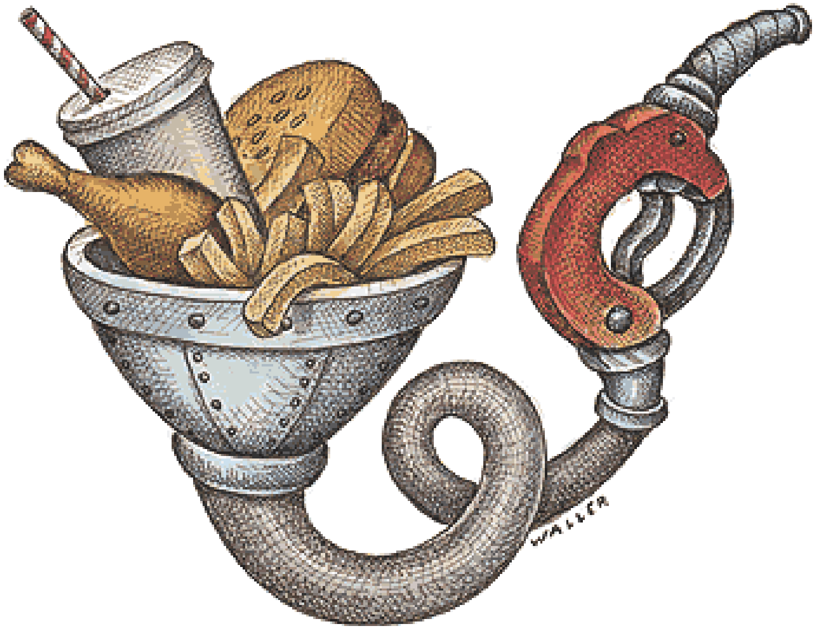
- **Potential energy** is energy that matter possesses because of its location or structure
- **Kinetic energy** is energy associated with motion
- **Thermal energy (heat)** is kinetic energy associated with random movement of atoms or molecules
- **Chemical energy** is potential energy available for release in a chemical reaction

A diver has more potential energy on the platform than in the water.

Diving converts potential energy to kinetic energy.

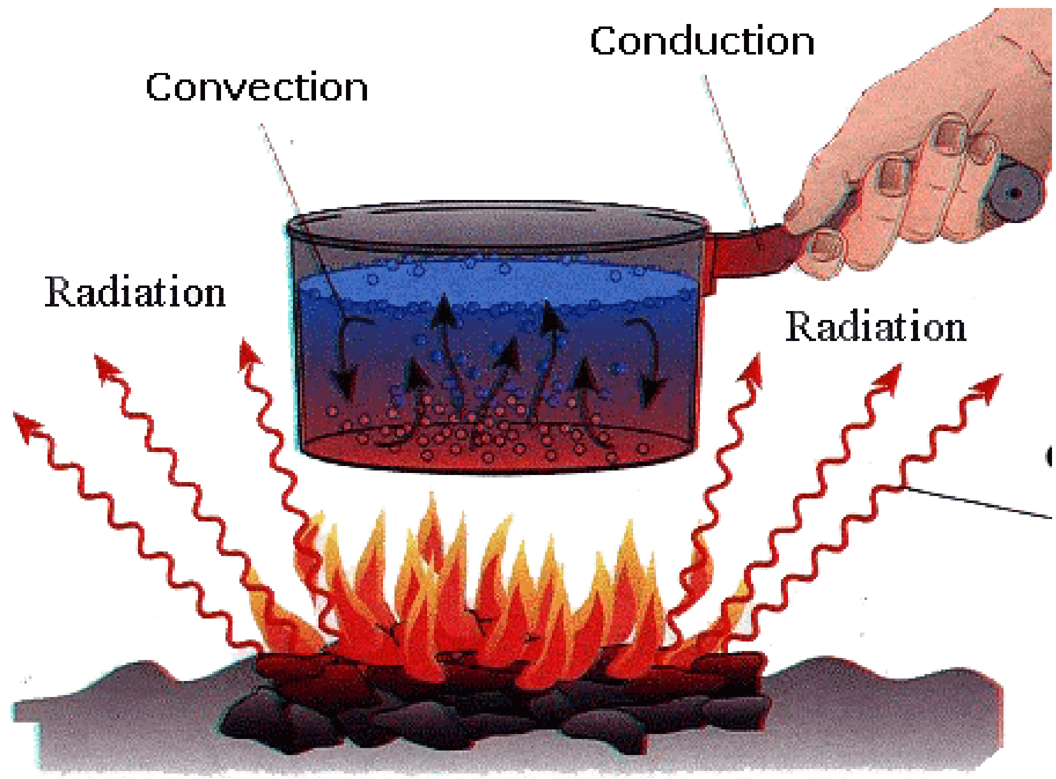
Climbing up converts the kinetic energy of muscle movement to potential energy.

A diver has less potential energy in the water than on the platform.



Chemical energy

Thermal energy



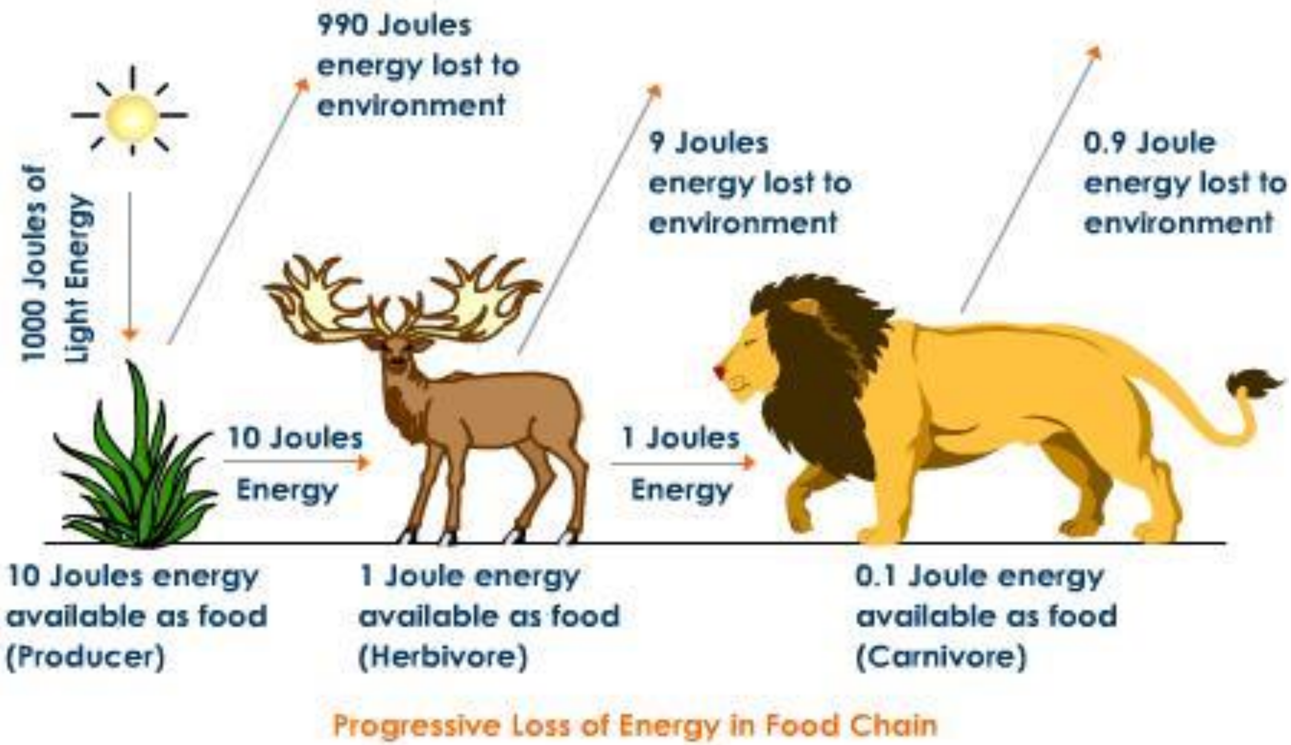
LAWS OF THERMODYNAMICS

- *Thermodynamics*: the study of energy transformation
- 2 laws that govern energy transformation:
  - First law of thermodynamics
  - Second law of thermodynamics

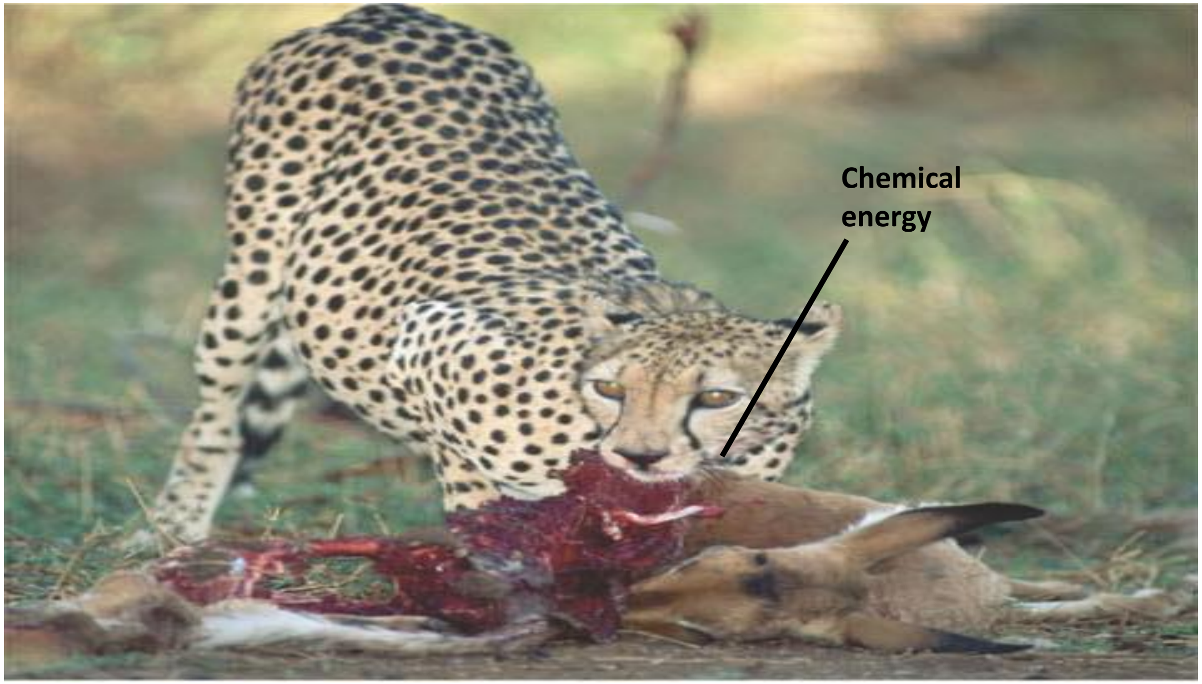
1^{st} Law of Thermodynamics

Energy can be transferred and transformed but it cannot be created or destroyed

- Principle of conservation of energy



1st Law of Thermodynamics



(a)

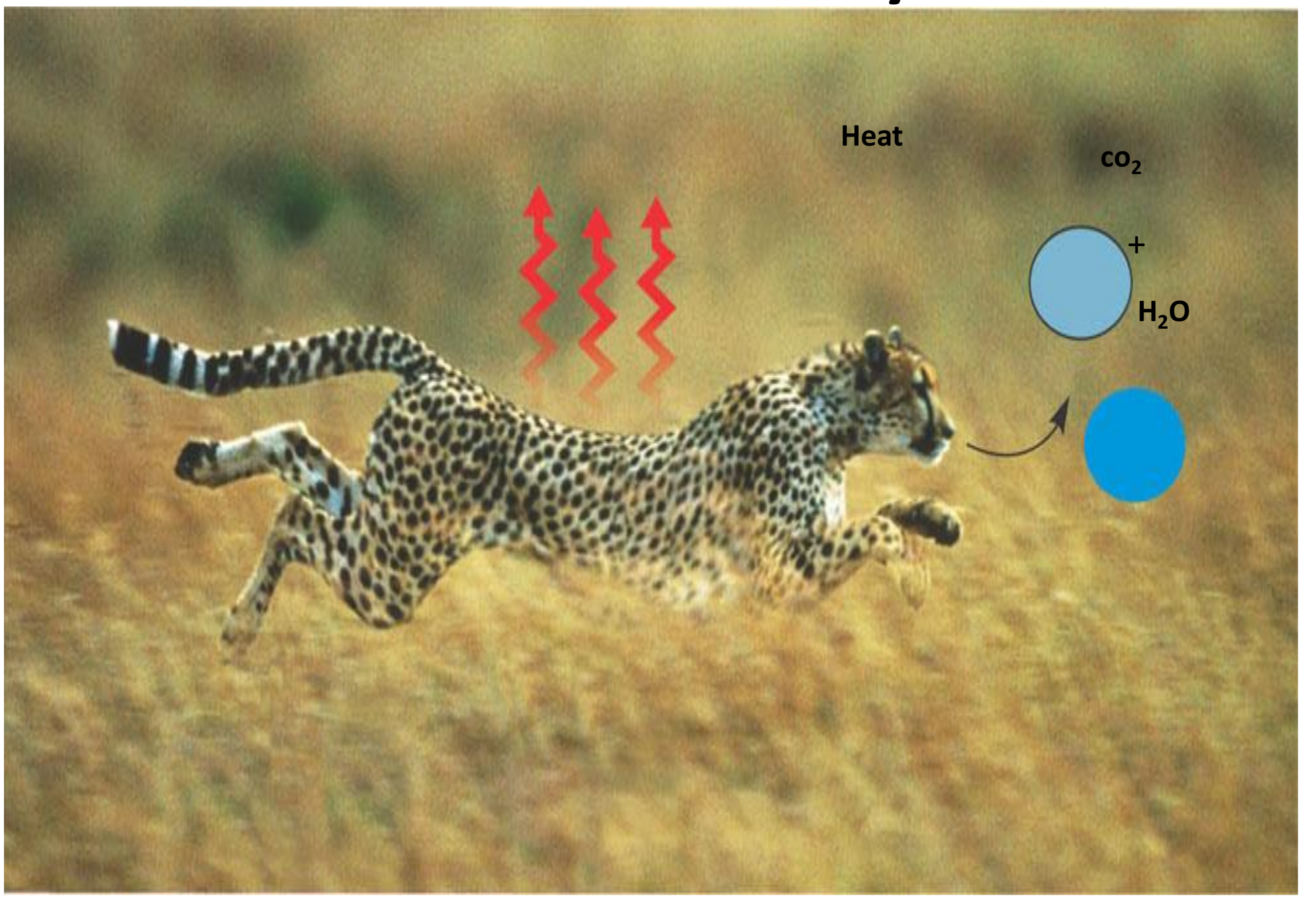
First law of thermodynamics: Energy can be transferred or transformed but neither created nor destroyed. For example, the chemical (potential) energy in food will be converted to the kinetic energy of the cheetah’s movement

2nd Law of Thermodynamics

Every energy transfer increase the entropy of the universe

- During every energy transfer or transformation, some energy is unusable, and is often lost as heat
- The more energy that is lost by a system to its surroundings, the less ordered and more random the system is; the measure of this randomness and disorder is known as **entropy**
- **Spontaneous process** that do not require energy input are energetically favorable and increase the entropy, or disorder, of the universe

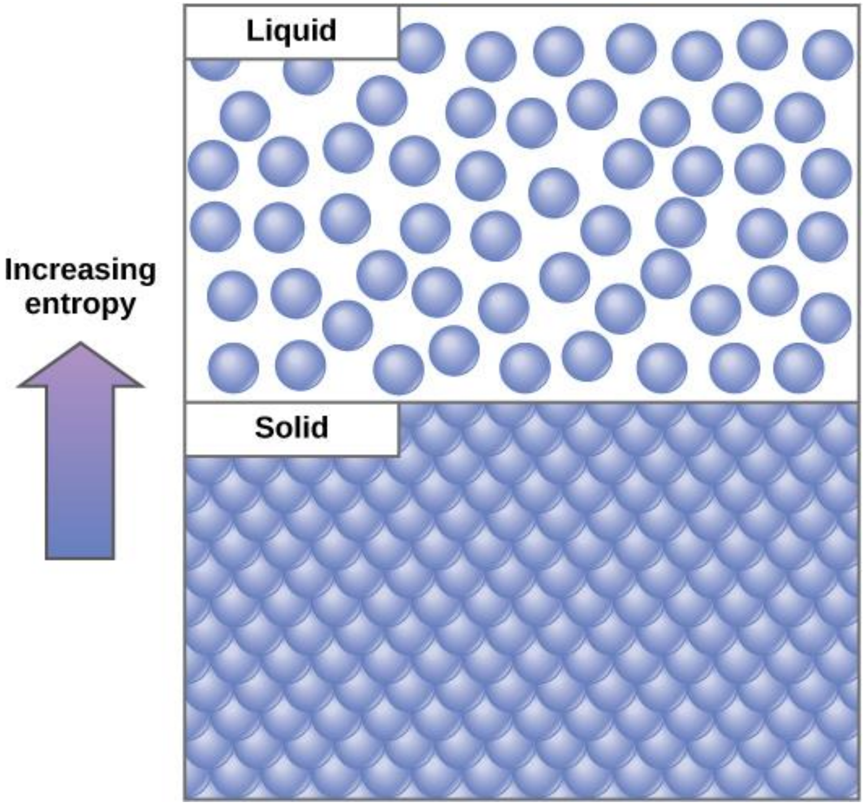
2nd Law of Thermodynamics



Second law of thermodynamics: Every energy transfer or transformation increases the disorder (entropy) of the universe. For example, disorder is added to the cheetah's surroundings in the form of heat and the small molecules that are the by-products of metabolism.

Entropy

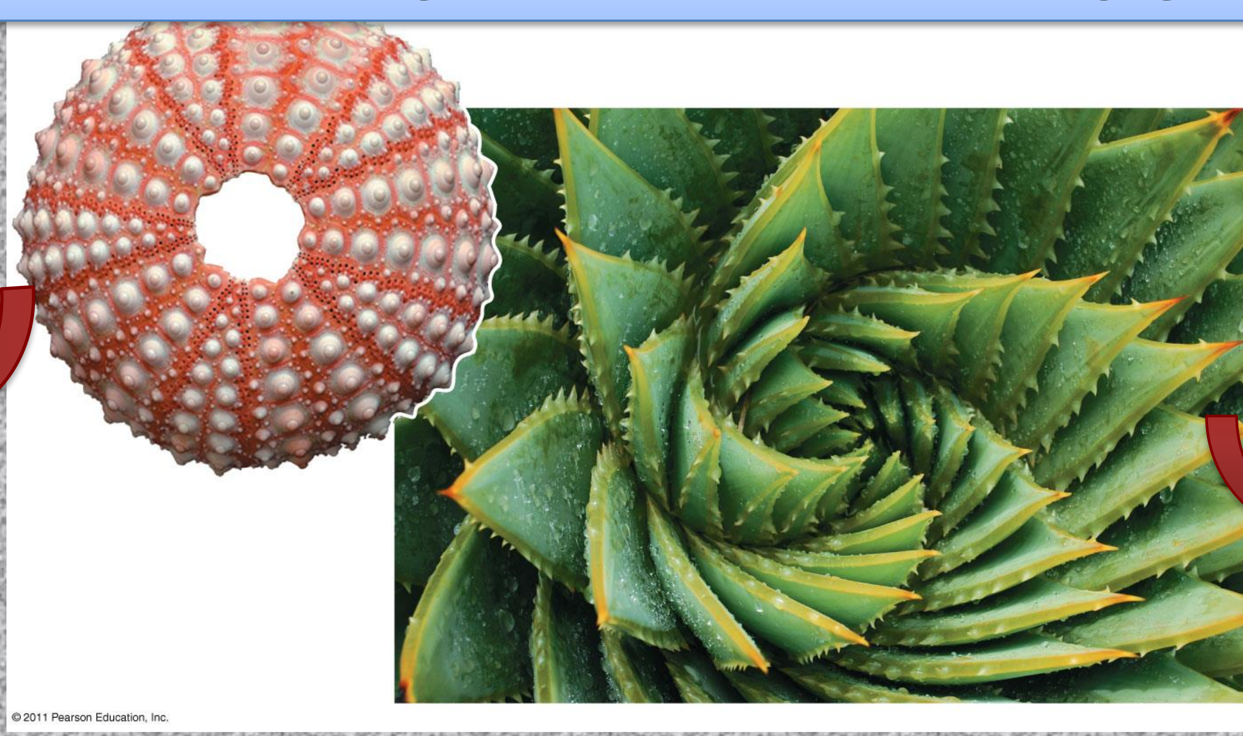
- Entropy: a measure of randomness or disorder in a system
- High entropy: high disorder and low energy



Order in the surrounding ↓ (entropy ↑)

Order in a system ↑ (entropy ↓)

CO_2



O_2

Figure 6.4

Measure of entropy: Gibb's Free Energy

- Gibb's free energy equation:

$$\Delta G = \Delta H - T\Delta S$$

- **G** = free energy
- **H** = enthalpy / heat content
- **T** = absolute temperature (in Kelvin)
- **S** = entropy/disorder

- Only processes with a negative Δ*G* are spontaneous
- Spontaneous processes can be harnessed to perform work

Free-Energy Change, ΔG

- A living system's free energy
  - Is energy that can do work under cellular conditions
- The change in free energy, ΔG during a biological process
  - Is related directly to the enthalpy (heat content) change (ΔH) and the change in entropy
  - Spontaneous process lowers the system's ΔG (-ve ΔG)

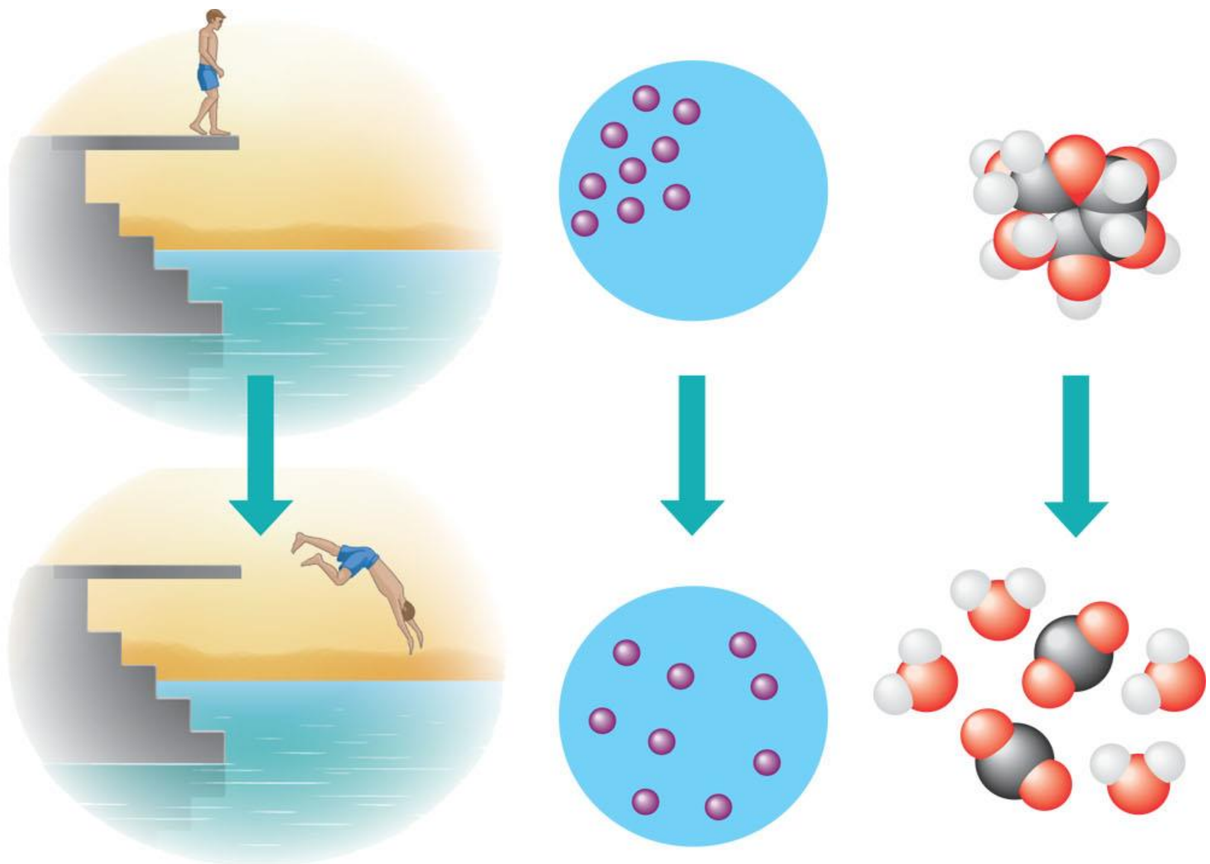
Free Energy, Stability, and Equilibrium

- Organisms live at the expense of free energy
- During a spontaneous process
  - Free energy decreases and the stability of a system increases
- At maximum stability
  - The system is at equilibrium
- A process is spontaneous and can perform work only when it is moving toward equilibrium

- More free energy (higher G)
- Less stable
- Greater work capacity

In a **spontaneously change**
- The free energy of the system decreases ($\Delta G<0$)
- The system becomes more stable
- The released free energy can be harnessed to do work

- Less free energy (lower G)
- More stable
- Less work capacity



(a) Gravitational motion. Objects move spontaneously from a higher altitude to a lower one.

(b) Diffusion. Molecules in a drop of dye diffuse until they are randomly dispersed.

(c) Chemical reaction. In a cell, a sugar molecule is broken down into simpler molecules.

Figure 6.5

Free energy and metabolism

- An **exergonic/exothermic** reaction
  - Proceeds with a net release of free energy and is spontaneous

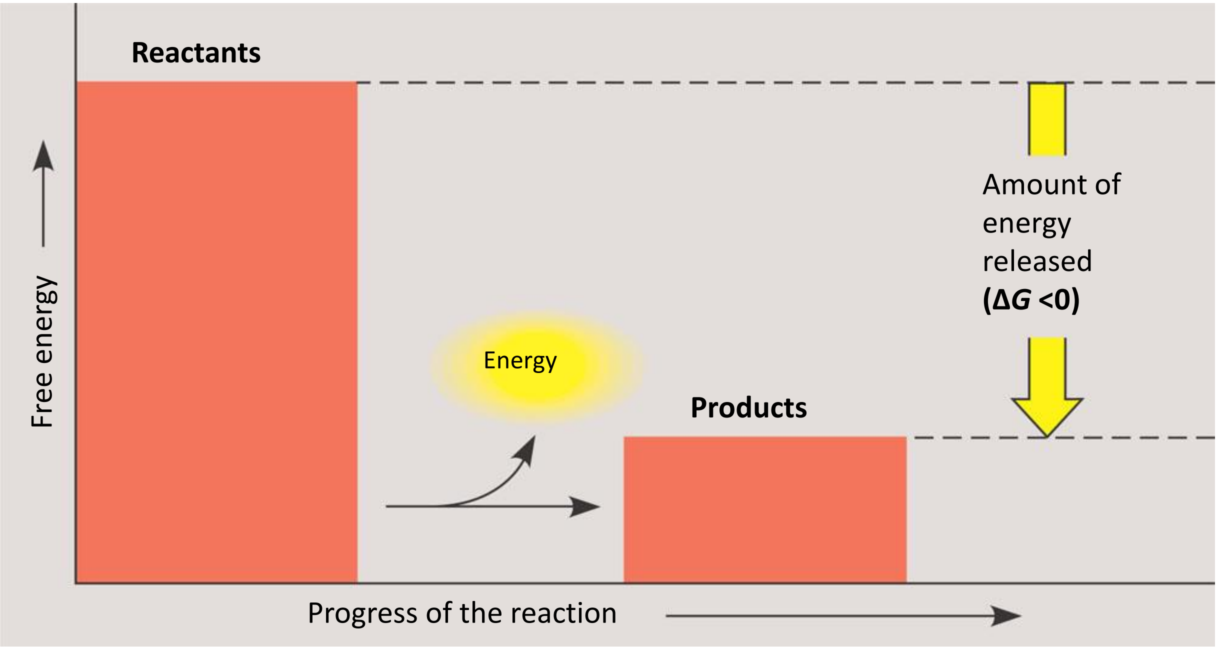


Figure 6.6

(a) Exergonic reaction: energy released

Free energy and metabolism

- An **endergonic/endothermic** reaction
  - Absorbs free energy from its surroundings and is nonspontaneous

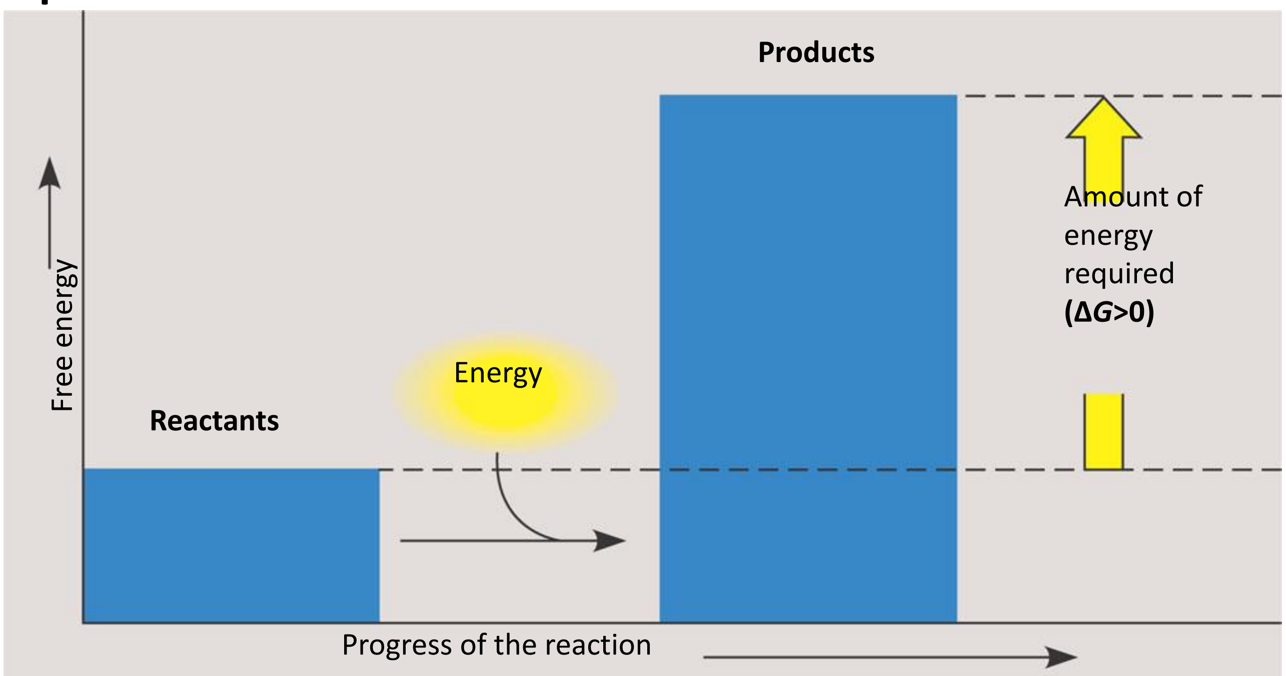
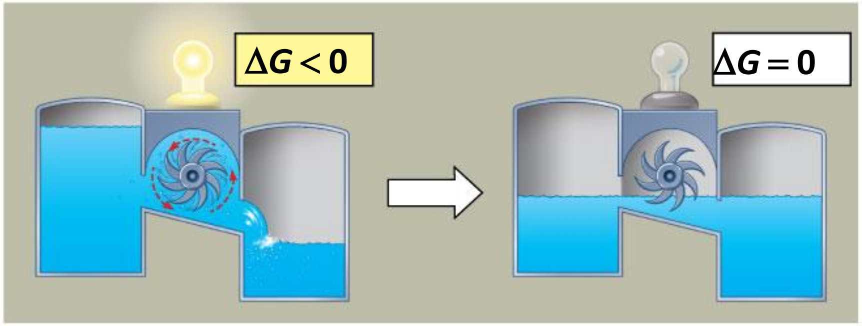


Figure 6.6

(b) Endergonic reaction: energy required

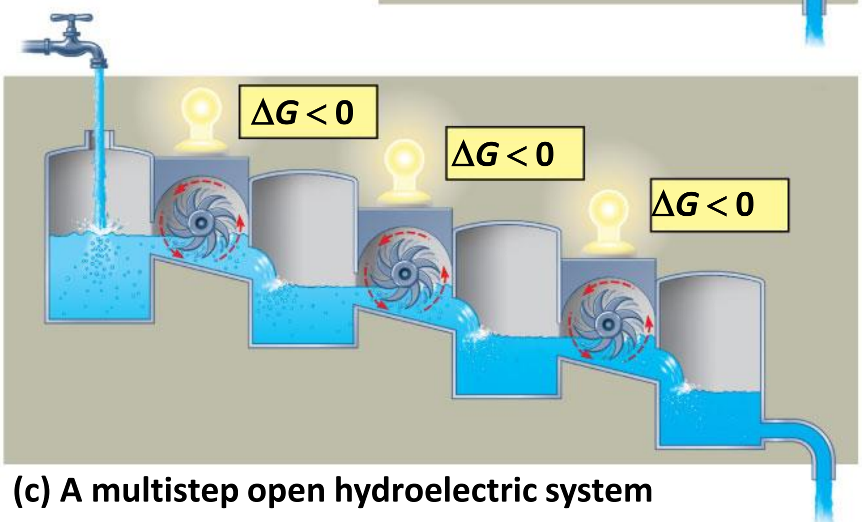
Metabolism

- Reactions in a closed system eventually reach equilibrium and then do no work
- The net direction of reactions in metabolism (forward or reverse) is determined by the concentration of the reactants and end products
- Chemical equilibrium is when the rate of the forward and reverse reactions are equal i.e. no net change
- Cells are not in equilibrium; they are open systems experiencing a constant flow of materials
- A defining feature of life is that metabolism is **never at equilibrium**
- A catabolic pathway in a cell releases free energy in a series of reactions
- Closed and open hydroelectric systems can serve as analogies

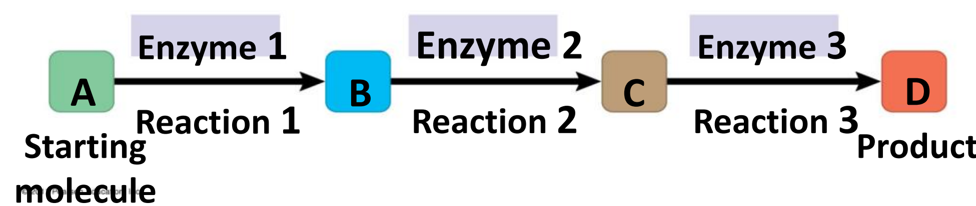


(a) An isolated hydroelectric system

(b) An open hydro-electric system



(c) A multistep open hydroelectric system



Videos to help you

- Entropy: Embrace the Chaos!

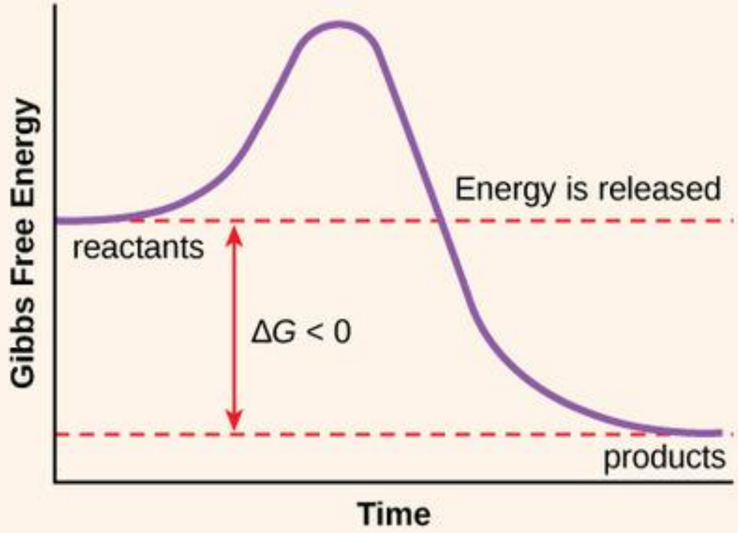
https://www.youtube.com/watch?v=ZsY4WcQOrfk

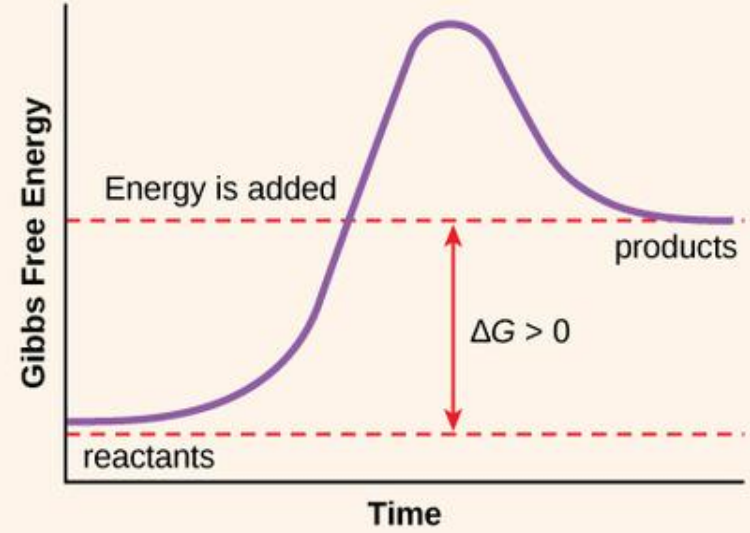
A *cool* demo at 4:28

- Gibb's Free Energy

https://www.youtube.com/watch?v=DPjMPeU5OeM

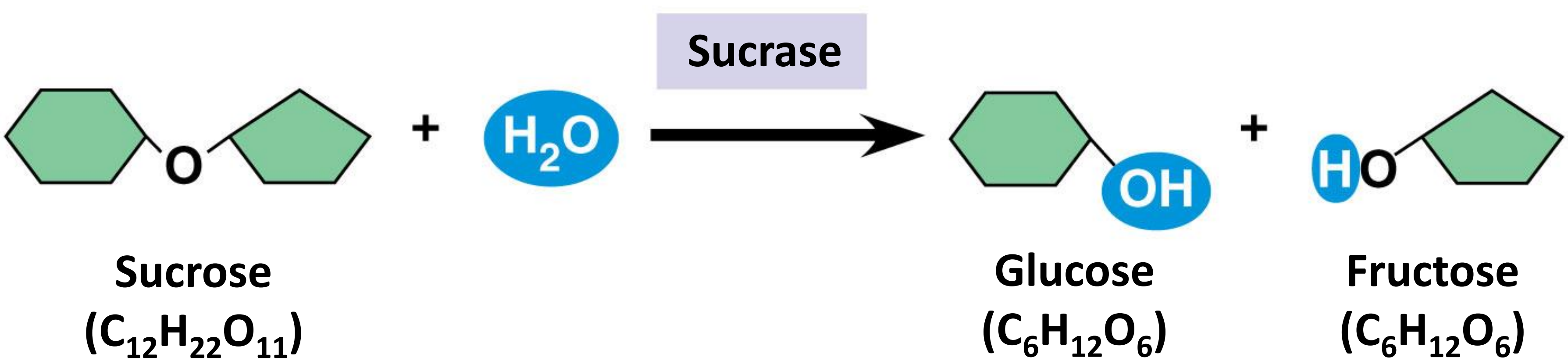
EXERGONIC REACTION: ΔG < O	ENDERGONIC REACTION: ΔG > O
Reaction is spontaneous	**Reaction is not spontaneous**





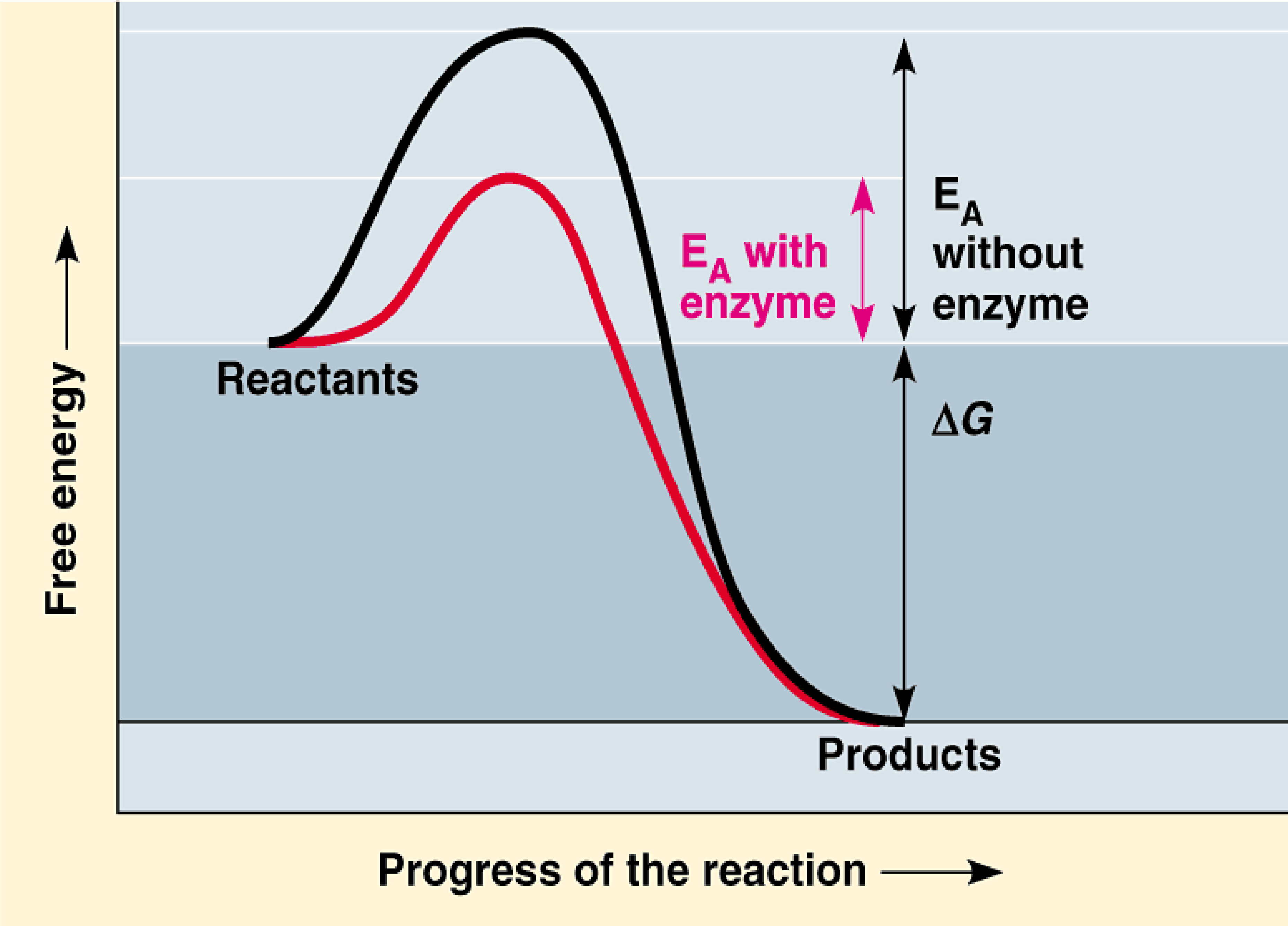
Enzymes

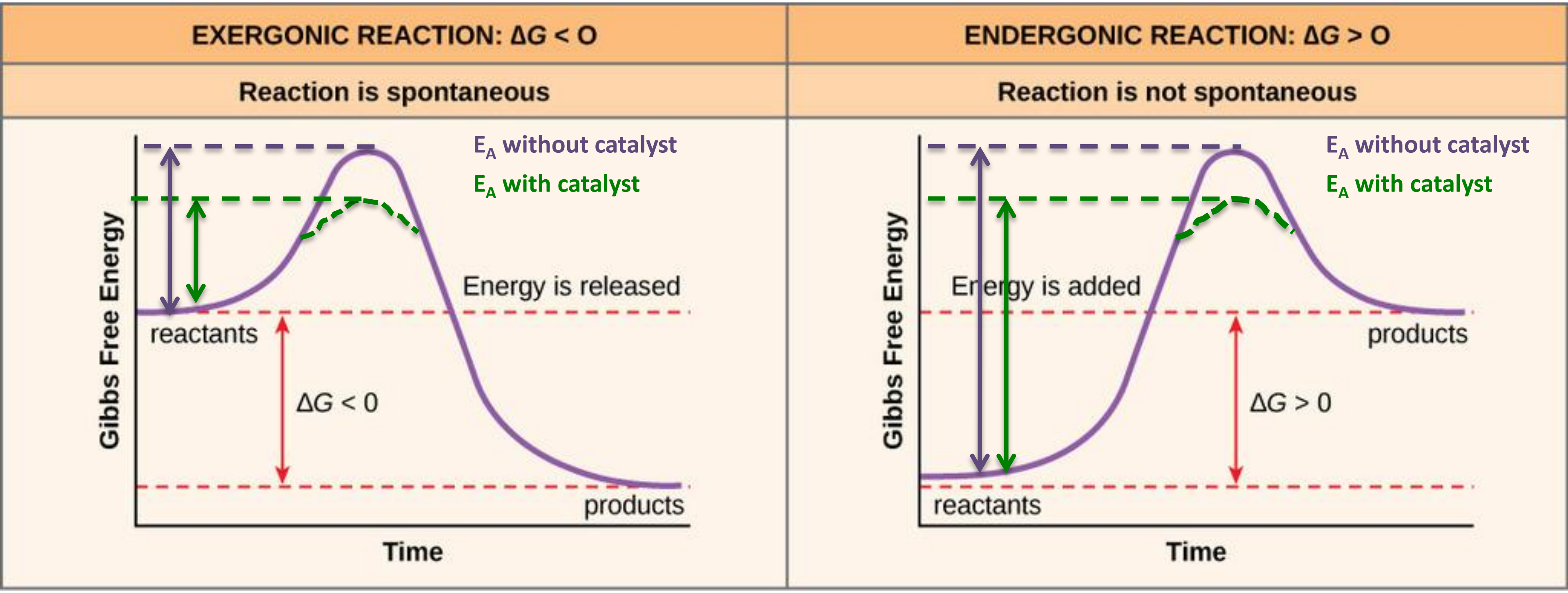
- Catalytic proteins (catalyst) speed up reactions by lowering activation energy E_A
- A catalyst
  - A chemical agent that speeds up a reaction without being consumed by the reaction



Enzymes lower activation energy barrier

- The **activation energy**, $\mathbf{E_A}$
  - Is the initial amount of energy needed to start a chemical reaction
  - Is often supplied in the form of heat from the surroundings in a system


Free energy
Reactants
E_A with enzyme
E_A without enzyme
ΔG
Products
Progress of the reaction



- E_A is lowered in an enzymatic reaction
- *ΔG* is unchanged

Characteristics of Enzymes

- Globular proteins exhibit as tertiary structure
- Substrate specific
- Remain unchanged in a reaction
- Can be reused
- Name after their substrate, wth suffix *-ase* at the end e.g. lactase, dehydrogenase
- Catalyze reactions in both directions
- Assisted by cofactors or coenzyme

Active Site

- The active site
  - Is the region on the enzyme where the substrate binds

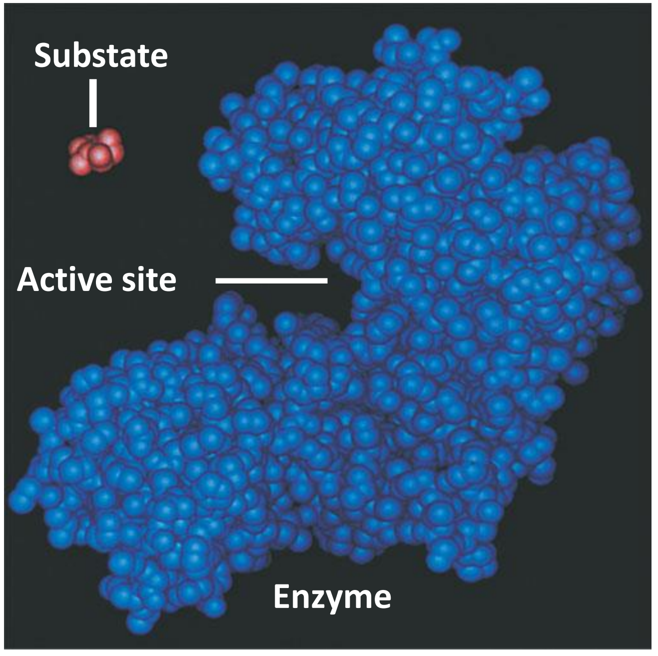
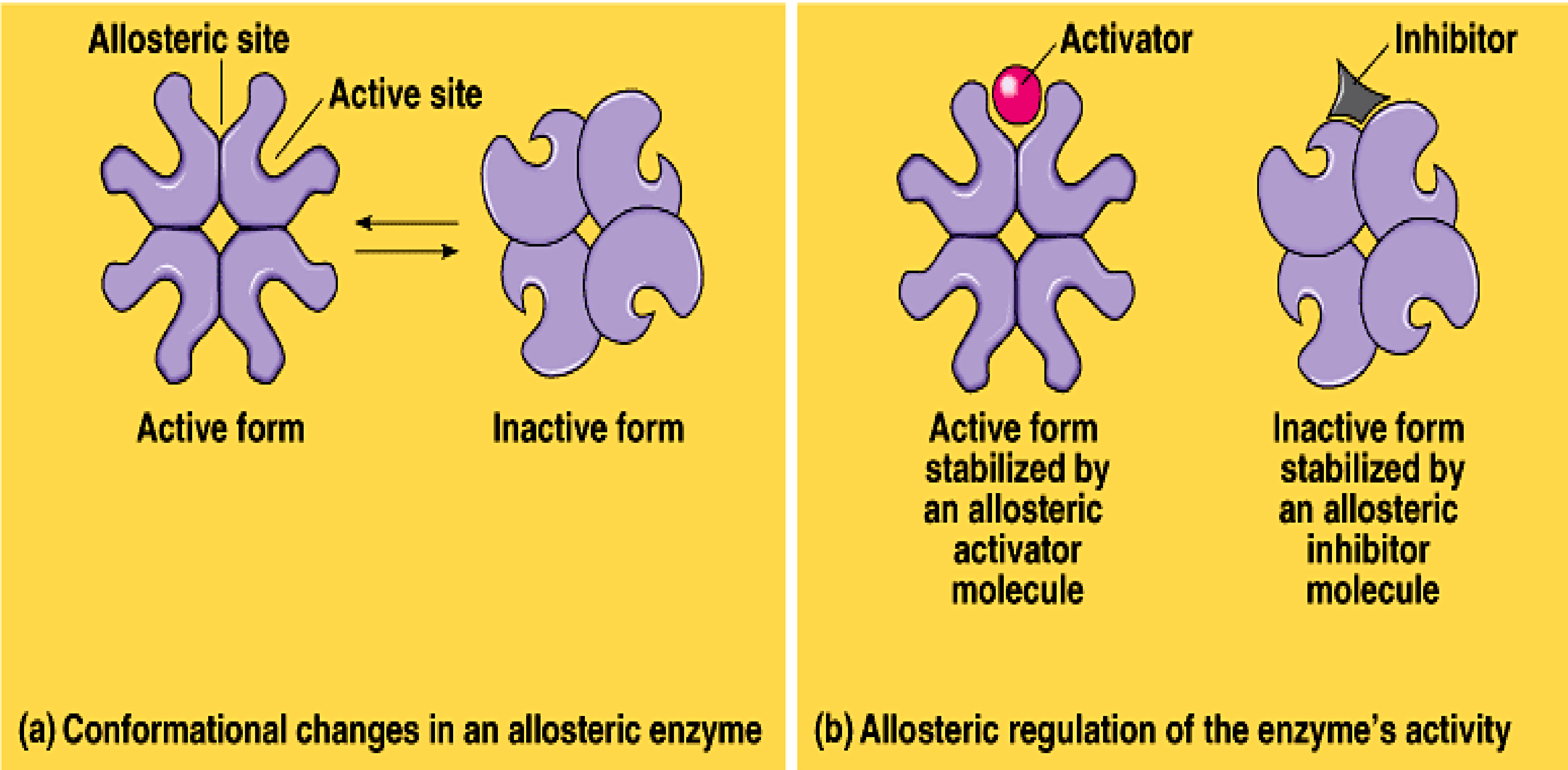


Figure 8.16

Allosteric Site

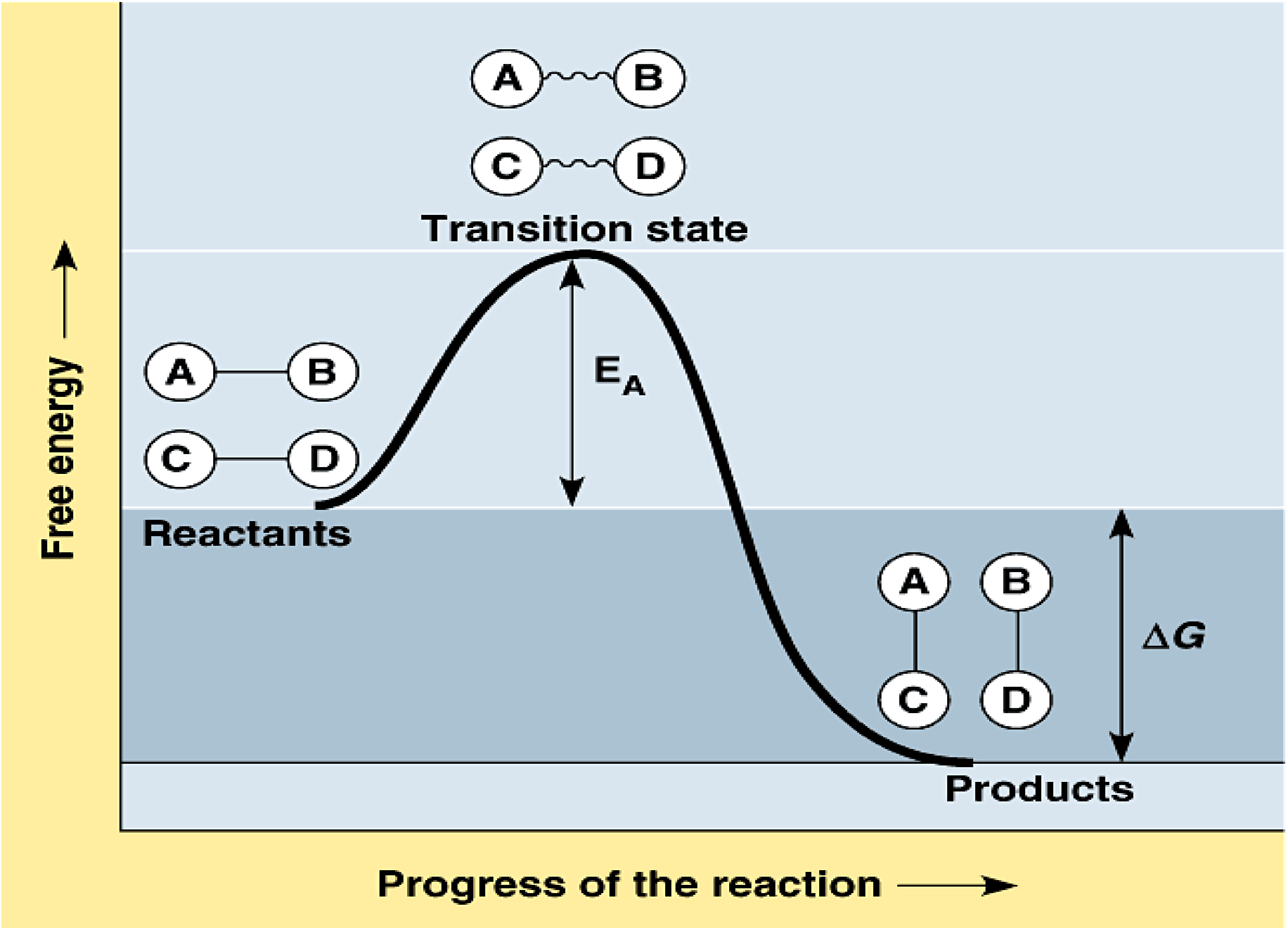
- ALLOSTERIC ENZYMES have two kinds of binding sites, one active site for the substrate, and an ALLOSTERIC SITE for an allosteric effector.

- There are 2 kinds of allosteric effectors:
  - Allosteric activator: induces active form of enzyme
  - Allosteric inhibitor: induces inactive form of enzyme

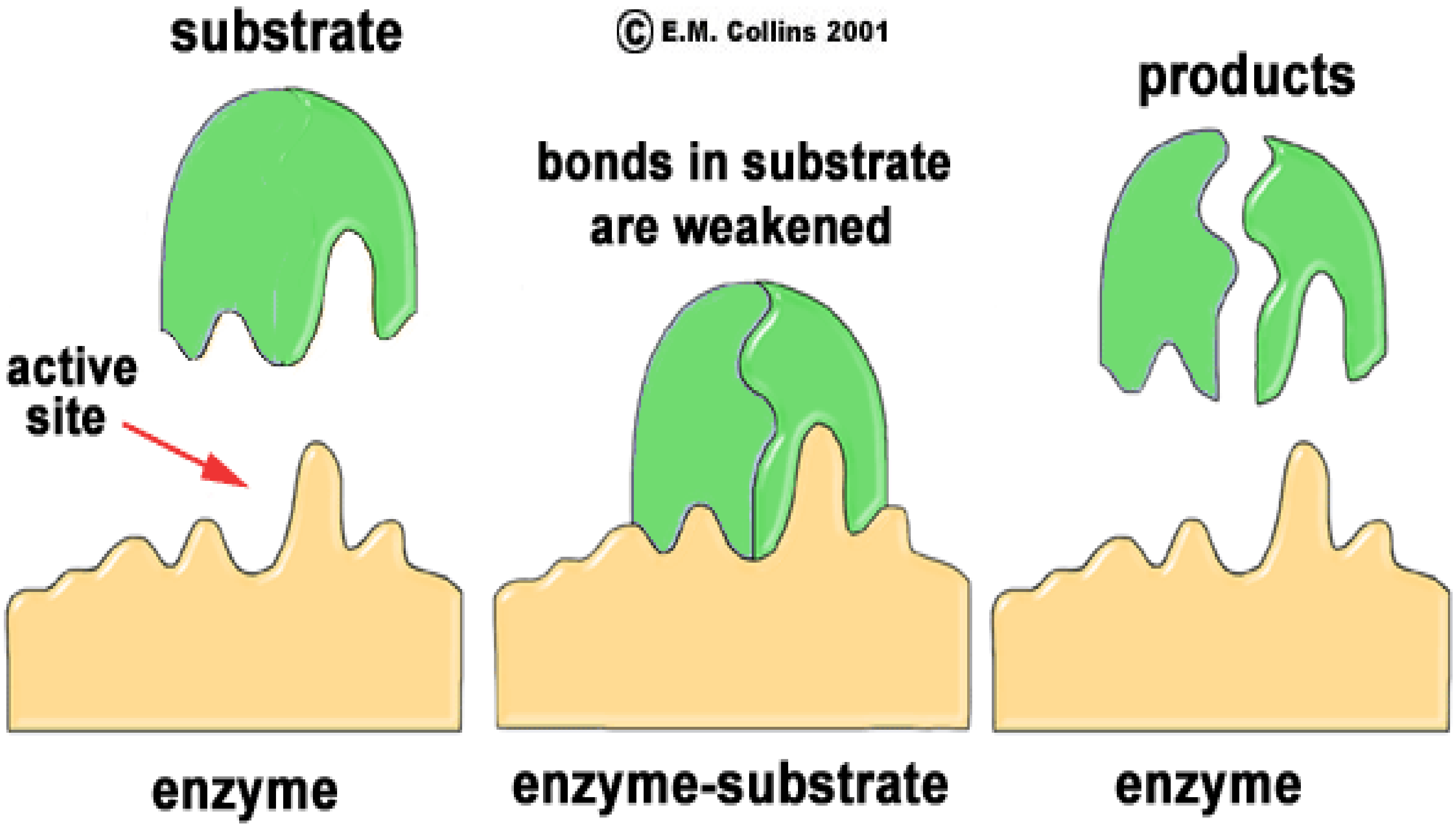

Allosteric site
Active site
Active form
Inactive form
Activator
Inhibitor
Active form stabilized by an allosteric activator molecule
Inactive form stabilized by an allosteric inhibitor molecule
(a) Conformational changes in an allosteric enzyme
(b) Allosteric regulation of the enzyme's activity

State of Reactions

- The action of enzymes catalyze metabolic reactions.
- There are three stages to these reactions
  - Reactant state
  - Transition state
  - Product state

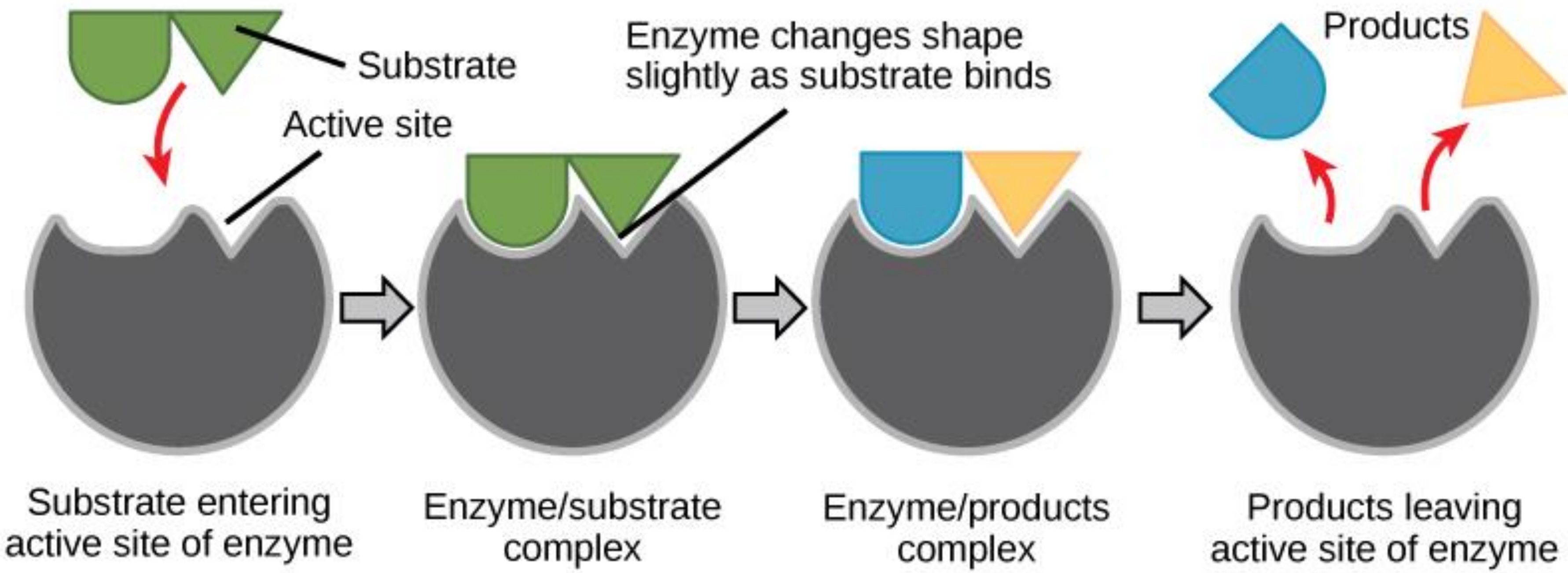

A
B
C
D
Transition state
Free energy
E$_A$
A
B
C
D
Reactants
A
B
C
D
ΔG
Products
Progress of the reaction

Catalytic cycle of enzyme



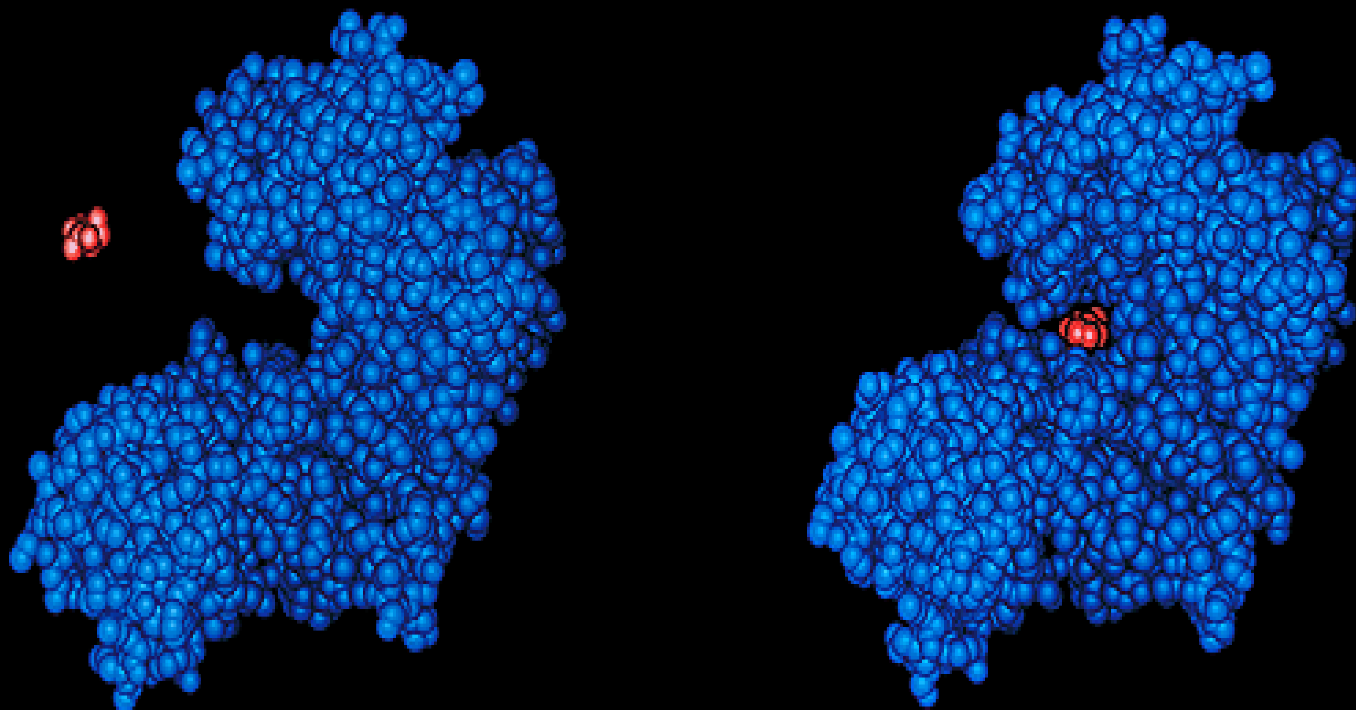
How Enzymes Work

- The **induced-fit model** describes how enzymes work.
- Within the protein (enzymes are protein) there is an active site with which the reactants interact.
- The interaction of substrate and enzyme cause the enzyme to change shape.
- The new position of the enzyme places the substrate molecules in a position favorable to their reaction. The product is released.



Induced Fit

According to the induced fit model, both enzyme and substrate undergo dynamic conformational changes upon binding. The enzyme contorts the substrate into its transition state, thereby increasing the rate of the reaction.

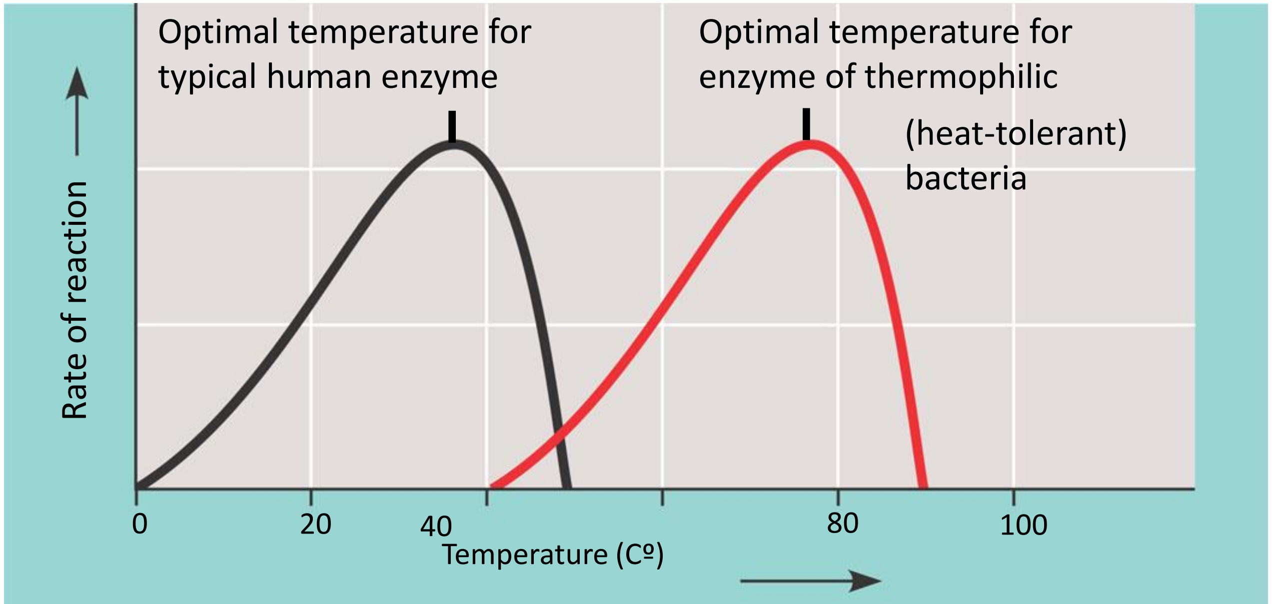


Effect of local condition on enzyme activity

- Temperature & pH
- Helpers molecules
  - Cofactor
    inorganic
  - Coenzyme
    organic
  - Prosthetic group
    Non-protein
- Enzyme inhibitor
  - Competitive inhibitor
  - Non-Competitive inhibitor

Enzyme Efficiency

- Temperature
  - Has an optimal temperature in which it can function the greatest (human’s 35-40°C)



(a) Optimal temperature for two enzymes

Figure 8.18

Thermophilic bacteria in a volcanic hotspring

Enzyme Efficiency

- pH
  – Has an optimal pH in which it can function

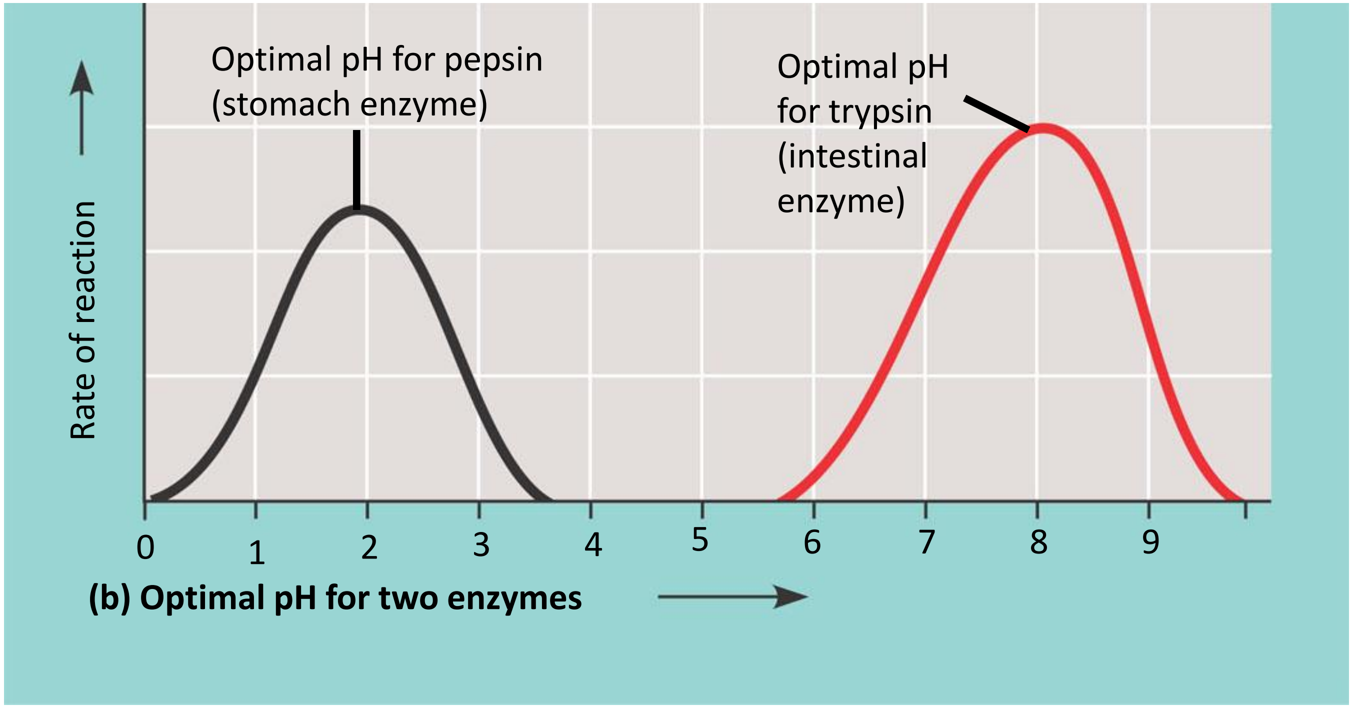
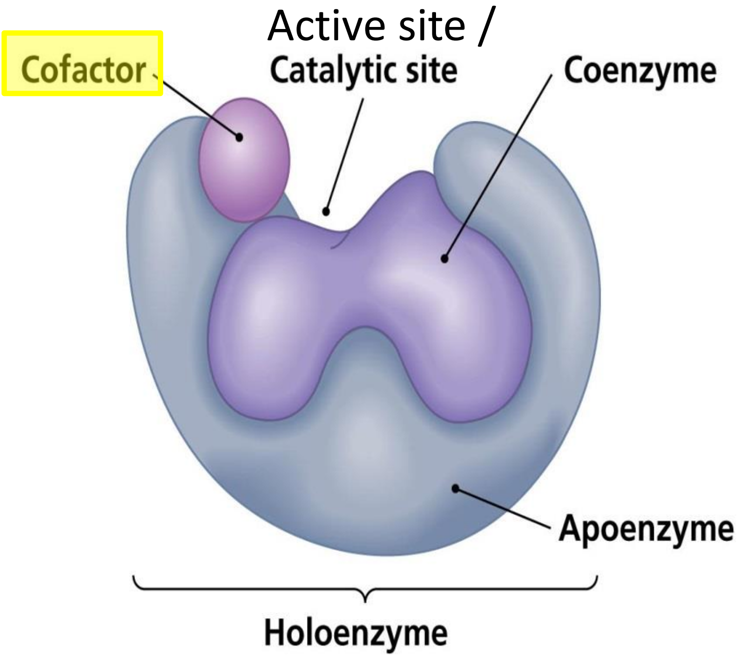


Figure 8.18

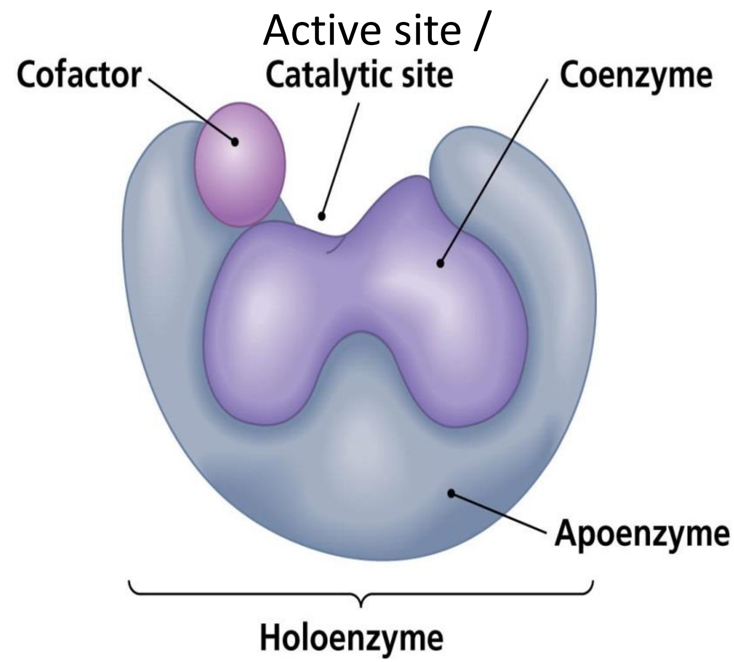
Cofactors

- Inorganic (Zn, Fe, Cu, Mg ions)
- Needed by certain enzyme to function efficiently
- Without it, enzyme function slowly or cannot work
- Classification
  - Coenzyme (organic co-factor)
  - Prosthetic group

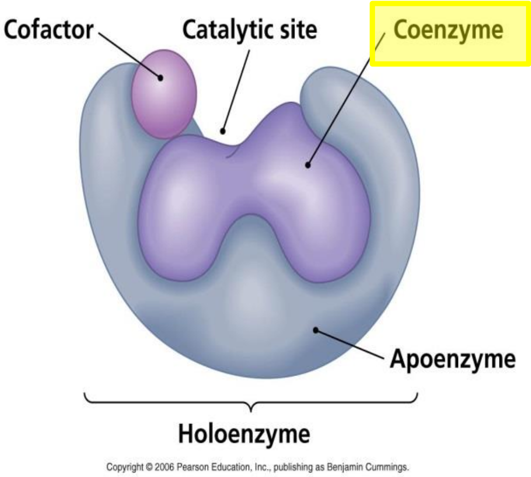


Properties of Cofactor:

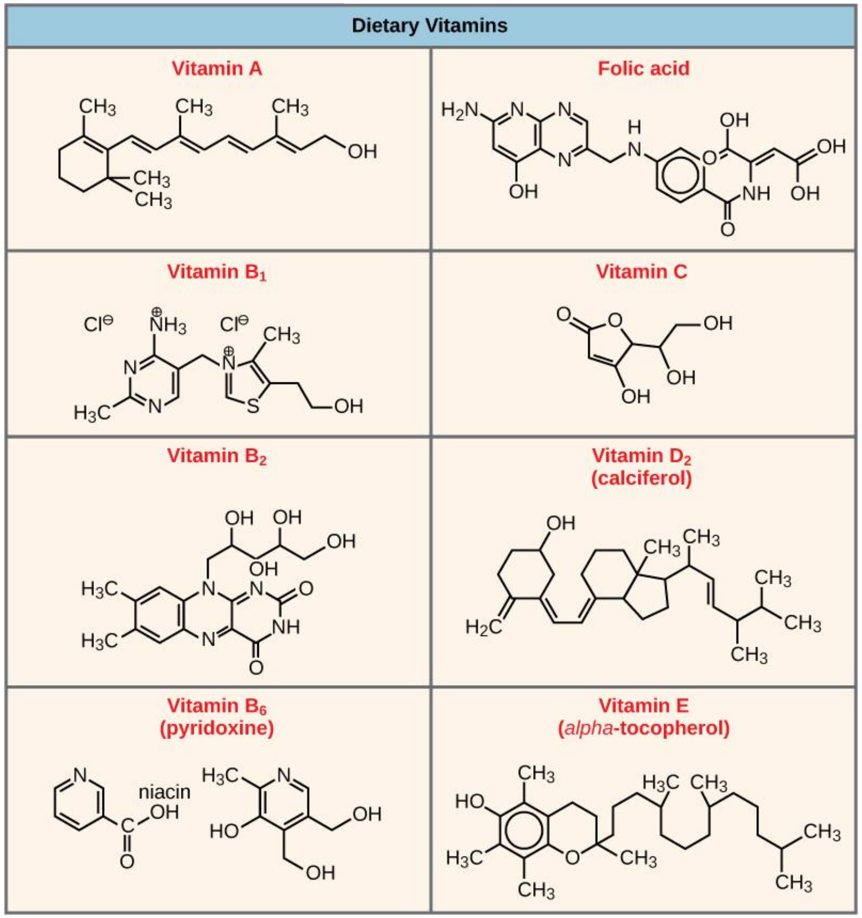
- Not denatured by high temperature
- Inorganic ion @ complex organic compound of medium size
- Regenerated at the end of the reaction or by another reaction later
- **Holoenzyme =** enzyme with cofactor
- **Apoenzyme** = enzyme without cofactor



Coenzyme

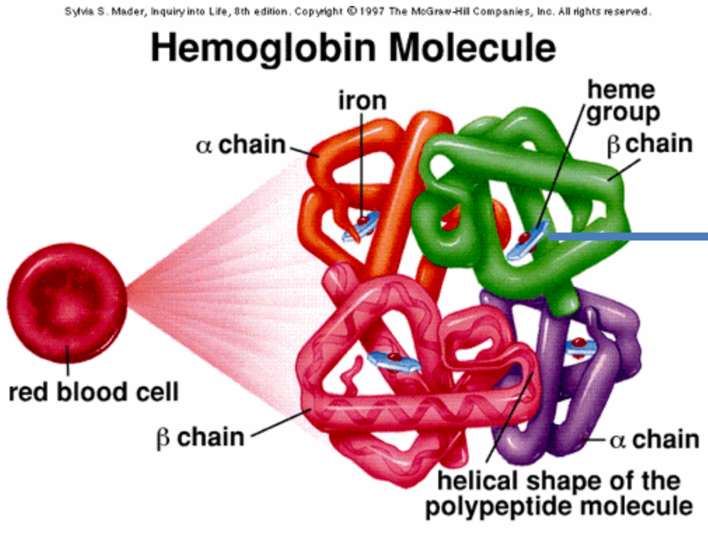


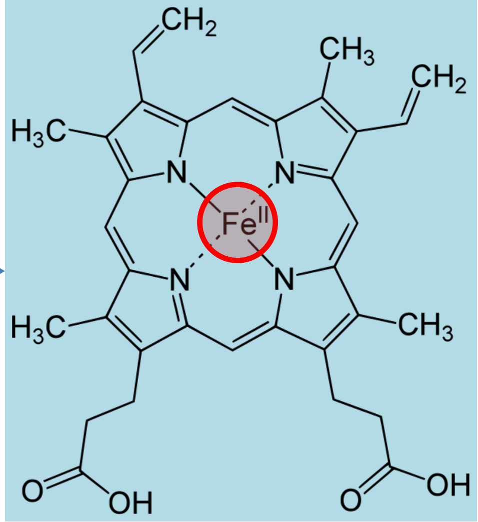
- Non-protein organic compound
- Most vitamins
- Eg: NAD (nicotinamide adenine dinucleotide) from vitamin niacin as electron acceptor or donor during cellular respiration



Prosthetic Group

- Non-protein organic compound bonded to enzyme/protein by **covalent bond** forming a conjugated protein.
- Eg: Haem derived from porphyrin contains iron (Fe^{2+}) in haemoglobin of RBC





Inhibitor

- Inorganic or organic compound that slows down or stops enzyme rxn.
- Chemicals that denature protein & other chemicals that can bind to a certain enzyme specifically.
- Can be:
  - Reversible → weak interaction
  - Irreversible (e.g. toxin and poison) → covalent bond

Competitive Inhibitors

- Bind to the active site of an enzyme, competing with the substrate

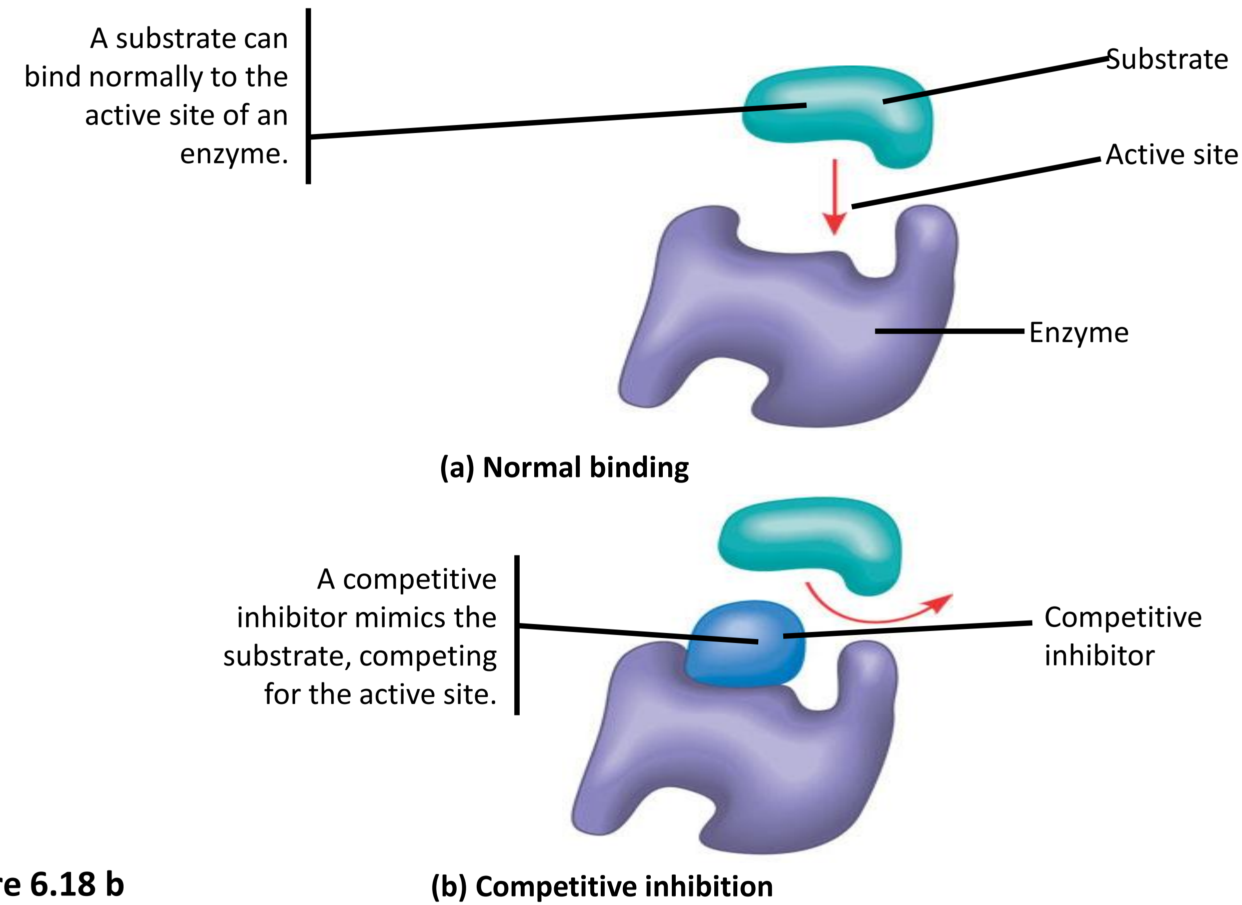


Figure 6.18 b

Noncompetitive Inhibitors

- Bind to another part of an enzyme (allosteric site), changing the function

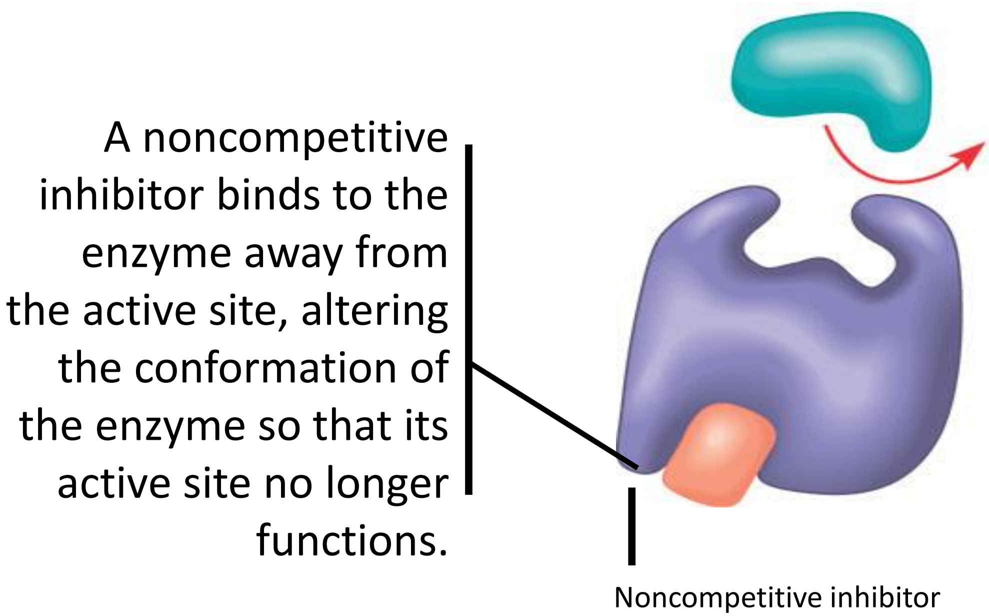
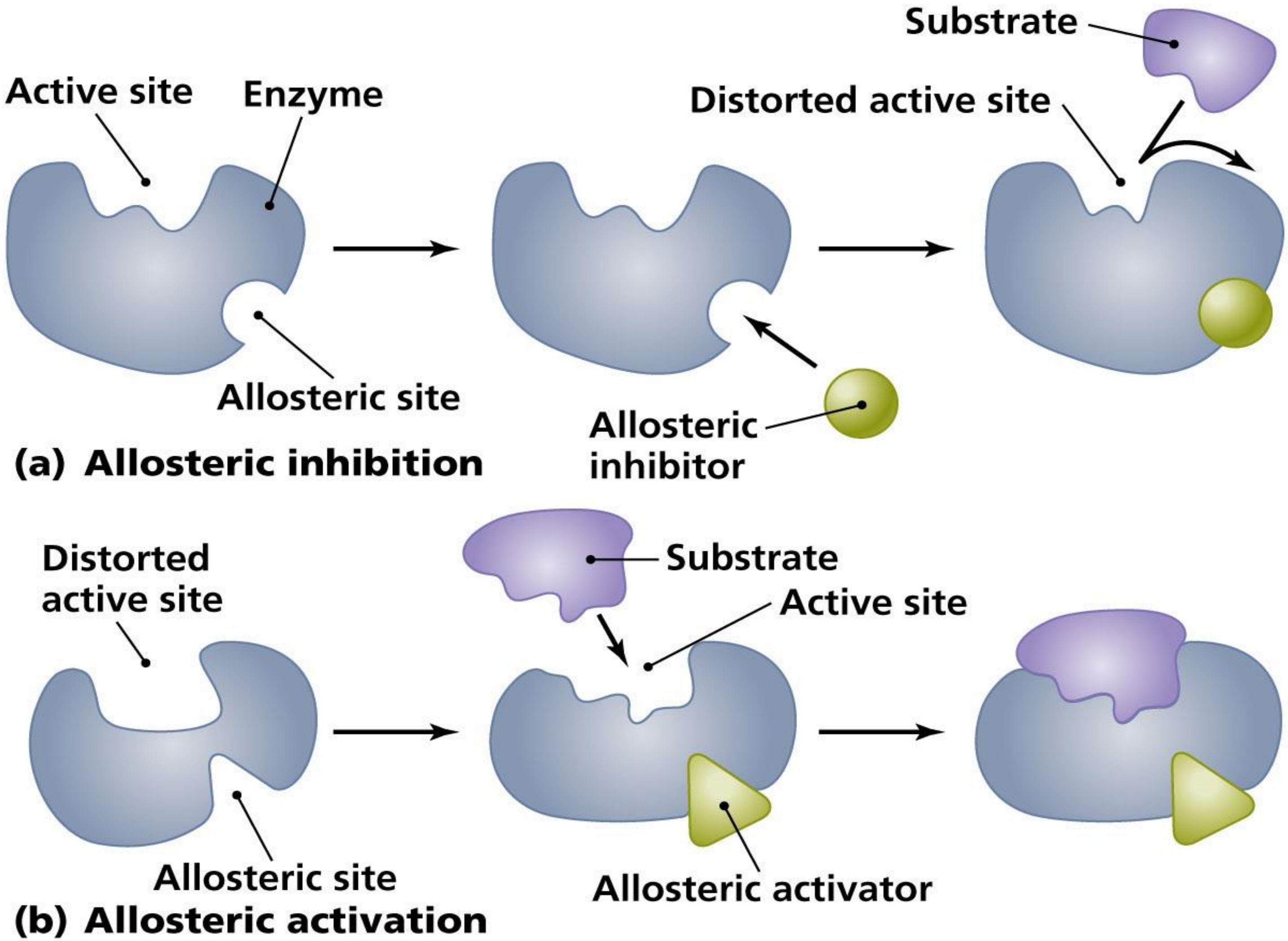


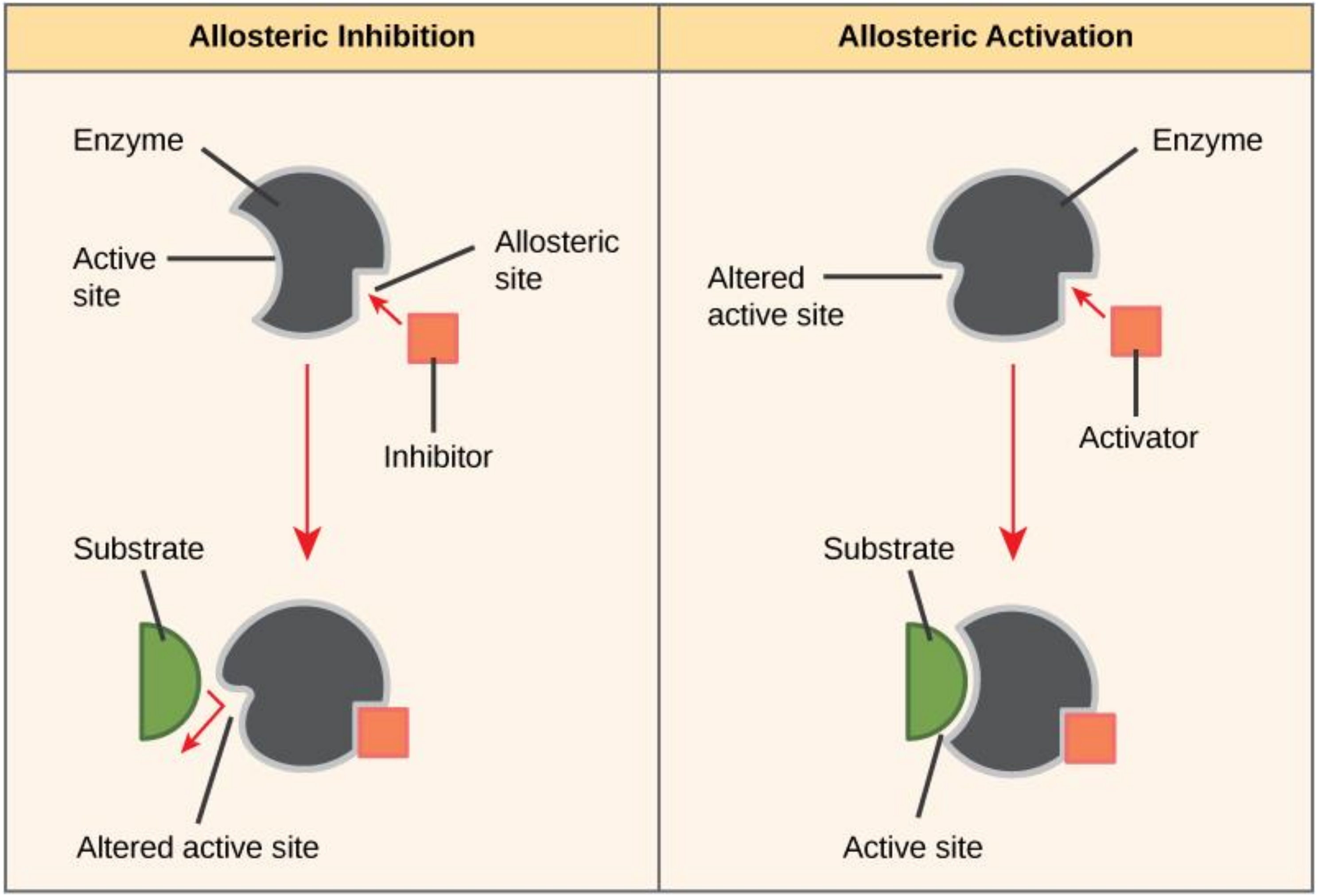
Figure 6.18 **(c) Noncompetitive inhibition**

Regulation of enzyme activity

- **Allosteric regulation** may either inhibit or stimulate an enzyme’s activity
- Allosteric regulation occurs when a regulatory molecule binds to a protein at one site and affects the protein’s function at another site (something like non-competitive inhibitor)

Allosteric inhibition and activation





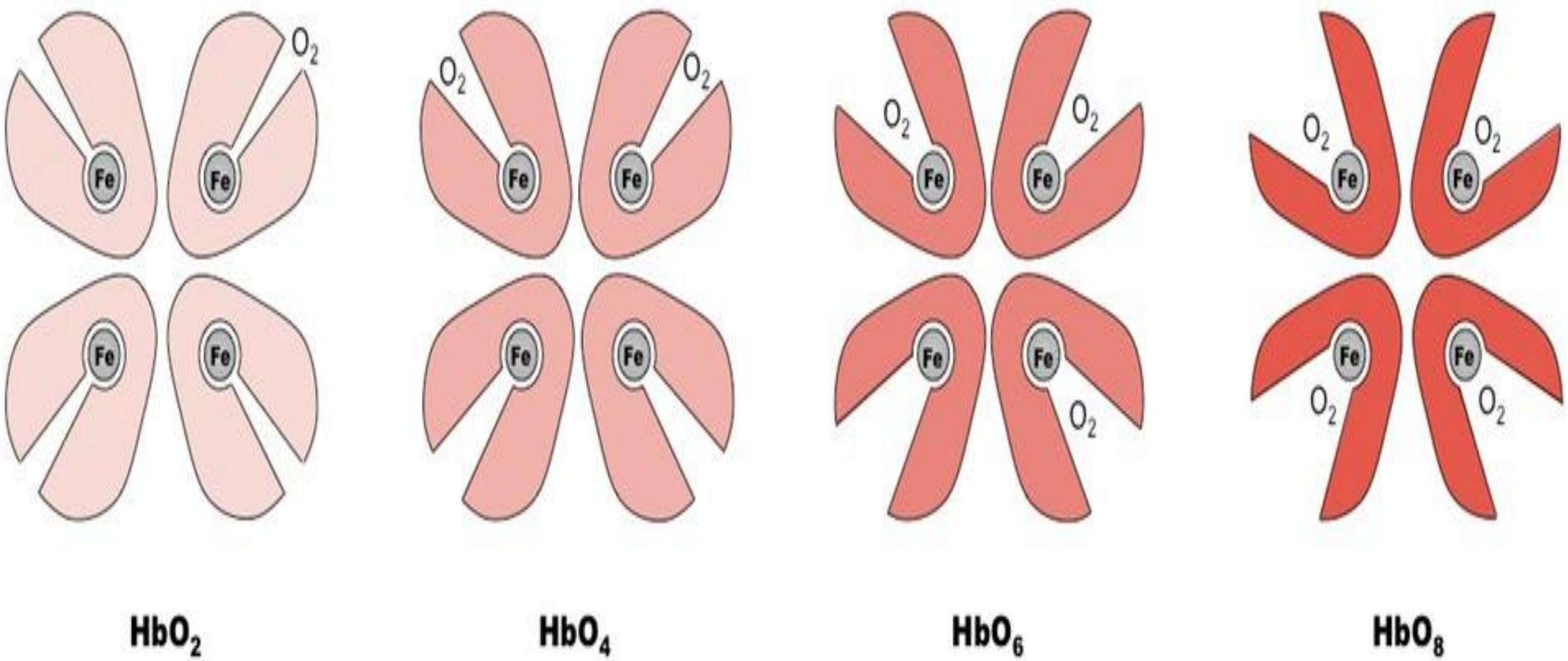
Allosteric Inhibitors and Activators

Allosteric inhibitors modify the active site of the enzyme so that substrate binding is reduced or prevented. In contrast, allosteric activators modify the active site of the enzyme so that the affinity for the substrate increases.

Cooperativity

- In cooperativity, an enzyme becomes more receptive to additional substrate molecules after one substrate molecule attaches to the active site.
- This occurs in quaternary structure, when enzymes are composed of multiple subunits (hemoglobin)

Cooperativity



Feedback inhibition

- Metabolic pathway inhibited by binding of an end product of an enzyme to an allosteric site on an enzyme
- Metabolism regulation

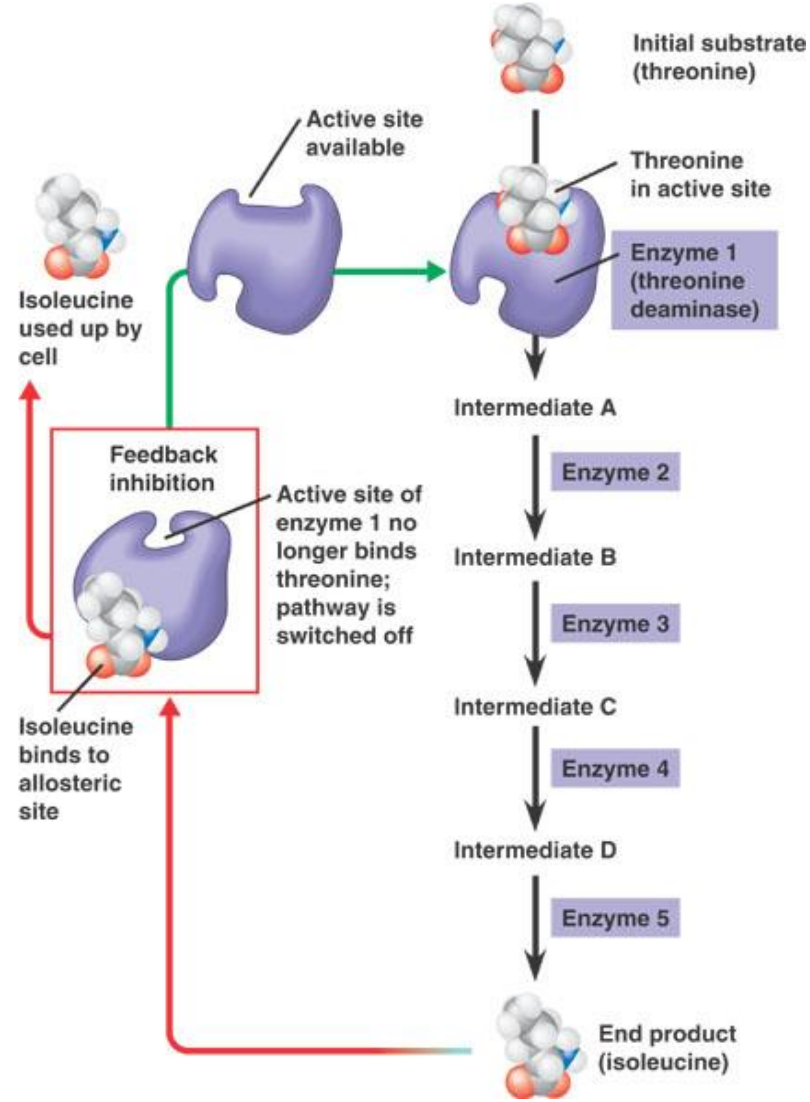


Figure 6.21

Classification of Enzymes

CLASSIFICATION OF ENZYMES		
Group of Enzyme	**Reaction Catalysed**	**Examples**
1. Oxidoreductases	Transfer of hydrogen and oxygen atoms or electrons from one substrate to another.	Dehydrogenases Oxidases
2. Transferases	Transfer of a specific group (a phosphate or methyl etc.) from one substrate to another.	Transaminase Kinases
3. Hydrolases	Hydrolysis of a substrate.	Estrases Digestive enzymes
4. Isomerases	Change of the molecular form of the substrate.	Phospho hexo isomerase, Fumarase
5. Lyases	Nonhydrolytic removal of a group or addition of a group to a substrate.	Decarboxylases Aldolases
6. Ligases (Synthetases)	Joining of two molecules by the formation of new bonds.	Citric acid synthetase

Thank you